

IVF / Fertility Clinic Management System

Hospital Administrator

Requirements Specification

Software Engineering – Laboratory Deliverable

Role: Armend (Administrator)

March 2026

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1. Administrator Role Overview

The Hospital Administrator (Armend) is the central operational role in the IVF / Fertility Clinic Management System. The administrator acts as the bridge between patients, doctors, lab technicians, and clinic management, ensuring that daily operations run smoothly.

Core Responsibilities:

- **Patient Management:** Register new patients, update their personal and insurance records, and maintain accurate patient directories.
- **Appointment Scheduling:** Create, modify, and cancel appointments across all doctors; resolve scheduling conflicts and prevent overbooking.
- **Room Assignment:** Assign consultation and procedure rooms to appointments, ensuring no double-bookings and optimal facility usage.
- **Doctor/Staff Management:** Add new doctors with their availability schedules, update profiles, and deactivate departed staff.
- **Service Catalog & Pricing:** Maintain the list of clinic services and their prices to ensure billing accuracy.
- **Billing & Payments:** Generate itemized patient bills, record payments, handle partial payments and insurance reimbursements.
- **Reporting:** Generate occupancy, revenue, and service utilization reports for clinic management decision-making.
- **Dashboard Monitoring:** Oversee daily clinic operations through a centralized dashboard showing appointments, room availability, pending bills, and recent registrations.

Access Level: The administrator has read/write access to patient records (non-clinical), appointment schedules, room management, billing, service catalog, and doctor profiles. The administrator does not have access to edit clinical medical records (doctor's domain) or laboratory results (lab technician's domain).

2. Functional Requirements

The following table lists all functional requirements specific to the Hospital Administrator role. Each requirement is traceable to one or more user stories.

Req #	Requirement	Priority	Comments / Traceability
FR_A01	The system shall provide a patient registration form that captures full name, date of birth, gender, national ID, contact details, insurance information, and medical history notes.	1	Supports US_A01. All mandatory fields must be validated before submission.
FR_A02	The system shall detect duplicate patient records based on national ID number and name + DOB combination and warn the administrator before allowing creation.	1	Prevents data integrity issues. Supports US_A01.
FR_A03	The system shall allow the administrator to search and filter the patient list by name, patient ID, appointment date, and treatment type with real-time results.	1	Search must return results after 2+ characters. Supports US_A10.
FR_A04	The system shall allow the administrator to update patient personal and insurance information, logging all changes with timestamps and administrator identity.	1	Previous values stored in audit history. Supports US_A11.
FR_A05	The system shall provide a doctor registration form capturing full name, specialization, license number, contact details, qualifications, and weekly availability schedule.	1	License number must be unique. Supports US_A04.
FR_A06	The system shall allow the administrator to deactivate a doctor, hiding them from new bookings while preserving their historical records and requiring reassignment of future appointments.	1	Supports US_A05. Deactivation cannot proceed until appointments are resolved.
FR_A07	The system shall display a calendar view of all doctors' appointment schedules with day, week, and month views, filterable by doctor, department, or date range.	1	Supports US_A07. Central scheduling tool for the administrator.
FR_A08	The system shall allow the administrator to create, reschedule, and cancel appointments on behalf of patients, with automatic conflict detection and notification to both patient and doctor.	1	Supports US_A07. Cancellation requires a mandatory reason.
FR_A09	The system shall display available rooms filtered by type and time window, and allow the administrator to assign a room to a confirmed appointment with conflict prevention.	2	Supports US_A02. Optimistic locking handles concurrent access.
FR_A10	The system shall provide a service catalog management interface allowing the administrator to add, edit, deactivate services, and set prices.	1	Supports US_A06. Service names must be

			unique within a category.
FR_A11	The system shall generate an itemized patient bill aggregating all unbilled services with automatic tax and insurance calculations, discount support, and invoice numbering.	1	Supports US_A03. Discounts require a mandatory reason.
FR_A12	The system shall allow the administrator to record payments against pending bills, supporting cash, card, bank transfer, and insurance methods with partial payment tracking.	1	Supports US_A08. Payment amount cannot exceed outstanding balance.
FR_A13	The system shall generate payment receipts upon successful payment recording and make them available to the patient in their portal.	1	Supports US_A08.
FR_A14	The system shall generate Occupancy, Revenue, and Service Utilization reports for a user-selected date range with optional filters and PDF export.	2	Supports US_A09. Reports are read-only.
FR_A15	The system shall display an administrator dashboard showing today's appointments, checked-in patients, available rooms, pending bill count/amount, and recent registrations.	2	Supports US_A12. Dashboard must load within 2 seconds.
FR_A16	The system shall send notifications (email/SMS) to patients and doctors upon appointment creation, rescheduling, cancellation, bill generation, and payment recording.	1	Notifications must be sent within 1 minute of the triggering event.
FR_A17	The system shall implement role-based access control, restricting administrator access to non-clinical patient data, scheduling, billing, and staff management.	1	Administrators cannot edit medical records or lab results.
FR_A18	The system shall maintain a complete audit trail of all administrator actions (patient registration, edits, billing, payments, schedule changes) with timestamps and user identity.	1	Required for regulatory compliance and dispute resolution.

3. Non-Functional Requirements

The following non-functional requirements define the quality attributes of the system as they apply to the Administrator's experience and the data the administrator manages.

Category	Req #	Requirement	Priority
Usability	NFR_A01	The administrator interface shall follow a consistent sidebar navigation pattern with clearly labeled sections (Dashboard, Patients, Appointments, Rooms, Doctors, Services, Billing, Reports).	1
Usability	NFR_A02	All administrator forms shall provide inline validation with descriptive error messages (e.g., 'National ID must be 10 characters') before submission.	1
Usability	NFR_A03	The system shall provide contextual help tooltips on complex fields during first-time use.	3
Performance	NFR_A04	The administrator dashboard shall load within 2 seconds under normal network conditions.	1
Performance	NFR_A05	Patient search results shall begin appearing within 500 milliseconds of the administrator typing 2 or more characters.	1
Performance	NFR_A06	Report generation shall complete within 10 seconds for date ranges up to 3 months.	2
Security	NFR_A07	Administrator passwords shall be stored using bcrypt hashing; plain-text storage is prohibited.	1
Security	NFR_A08	The system shall lock an administrator account after 5 consecutive failed login attempts.	1
Security	NFR_A09	Authentication shall use JWT tokens with a maximum session duration of 8 hours, requiring re-authentication after expiry.	1
Security	NFR_A10	All administrator actions shall be logged in an immutable audit trail including user ID, action, timestamp, and affected record.	1
Availability	NFR_A11	The system shall be available 99.5% of the time (approximately 44 hours maximum downtime per year), excluding scheduled maintenance windows.	1
Availability	NFR_A12	Automated daily backups shall be performed, with the ability to restore data to any point within the last 30 days.	1
Portability	NFR_A13	The administrator interface shall be a responsive web application compatible with Chrome, Firefox, Safari, and Edge on desktop and tablet devices.	1

Portability	NFR_A14	The interface shall adapt to screen sizes from 768px (tablet) to 1920px (desktop) without loss of functionality.	2
Data Protection	NFR_A15	Patient personal data shall be handled in compliance with Albanian Law No. 9887 on Protection of Personal Data and GDPR principles.	1
Data Protection	NFR_A16	Financial records (bills, payments, invoices) shall be retained for a minimum of 5 years as required by local regulations.	1
Maintainability	NFR_A17	The system shall support scheduled maintenance windows with advance notification to administrators, during which read-only access is maintained.	2
Scalability	NFR_A18	The system shall support at least 10 concurrent administrator sessions without performance degradation.	2

4. Administrator User Stories

This section contains 12 user stories for the Administrator role: 3 refined from the original requirements and 9 newly identified during analysis. Each story includes acceptance criteria that define when the story is considered complete.

Priorities: High = must-have for MVP (Sprint 1–2); Medium = important for Sprint 3–4.

ID	User Story	Acceptance Criteria	Pri.	Status
US_A01	As an administrator, I want to register a new patient by entering their personal details, contact information, and insurance data so that the patient record is created and they can begin receiving clinic services.	Patient record is created with a unique ID. All mandatory fields (full name, DOB, contact number, ID document) are validated. Duplicate patient detection warns the administrator before saving. Patient receives a welcome notification with portal access credentials.	High	Refined
US_A02	As an administrator, I want to assign a consultation or procedure room to a patient's appointment so that facility usage is organized and double-bookings are prevented.	Only rooms marked as 'Available' for the selected time slot can be assigned. Assigning a room changes its status to 'Occupied' for that time window. The system prevents assigning the same room to overlapping appointments. Room assignment is visible to the doctor and patient.	Medium	Refined
US_A03	As an administrator, I want to generate an itemized patient bill covering all services rendered (consultations, lab tests, procedures, medications, room charges) so that the clinic can collect accurate payments.	Bill includes all unbilled services with individual line items and prices. Taxes and insurance deductions are calculated automatically. Administrator can apply discounts with a mandatory reason field. Generated bill is visible to the patient in their portal. Bill status is set to 'Pending' until payment is recorded.	High	Refined

US_A04	As an administrator, I want to add a new doctor to the system by entering their professional details, specialization, and availability schedule so that patients can book appointments with them.	Doctor profile is created with name, specialization, license number, and contact. Weekly availability schedule is configured with time slots. Doctor appears in the appointment booking list immediately. System validates that the license number is unique.	High	New
US_A05	As an administrator, I want to update or deactivate an existing doctor's profile so that staffing records remain accurate and departed doctors no longer appear for new bookings.	Administrator can edit doctor details (contact, specialization, schedule). Deactivating a doctor hides them from new booking options. Existing future appointments for a deactivated doctor trigger a reassignment alert. Deactivated doctors remain in historical records for audit purposes.	High	New
US_A06	As an administrator, I want to manage the clinic's service catalog (add, edit, remove services and set prices) so that billing reflects the current offerings and pricing.	Administrator can add a new service with name, category, description, and price. Administrator can update the price of an existing service. Removing a service marks it as inactive rather than deleting it. Price changes take effect immediately for new bills but do not alter existing invoices.	High	New
US_A07	As an administrator, I want to manage appointment schedules across all doctors so that I can resolve conflicts, reschedule on behalf of patients, and prevent overbooking.	Administrator can view a calendar view of all doctors' schedules. Administrator can create, modify, or cancel appointments on behalf of patients. System prevents overlapping bookings for the same doctor.	High	New

		Patient and doctor are notified of any changes made by the administrator.		
US_A08	As an administrator, I want to record a payment against a patient's bill so that the financial records are updated and the patient's balance is accurate.	Administrator can select a pending bill and record payment amount and method. Partial payments are supported; remaining balance is shown. Bill status updates to 'Paid' when fully settled or 'Partially Paid' otherwise. A payment receipt is generated and available to the patient.	High	New
US_A09	As an administrator, I want to generate occupancy and revenue reports for a selected date range so that clinic management can monitor performance and make informed decisions.	Administrator can select report type and date range. Occupancy report shows room usage percentages and peak times. Revenue report shows income breakdown by service category. Reports can be exported as PDF.	Medium	New
US_A10	As an administrator, I want to search and filter the patient list by name, ID, appointment date, or treatment type so that I can quickly locate a specific patient's record.	Search returns results as the administrator types (minimum 2 characters). Filters can be combined. Results display patient name, ID, contact, and next appointment. Clicking a result opens the full patient profile.	High	New
US_A11	As an administrator, I want to update a patient's personal or insurance information so that their records remain current for billing and communication.	Administrator can edit patient contact details, address, and insurance information. Changes are logged with a timestamp and the administrator's identity. Previous values are preserved in an audit history. Updated insurance details are reflected in future bill calculations.	Medium	New

US_A12	As an administrator, I want to view a dashboard showing today's appointments, pending bills, room availability, and recent registrations so that I have an overview of daily clinic operations.	Dashboard loads within 2 seconds. Shows count of today's appointments, checked-in patients, and available rooms. Pending bills section shows total outstanding amount and count. Recent registrations list shows patients added in the last 7 days.	Medium	New
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5. User Scenarios List

The following table summarizes all user scenarios for the Administrator role. Each scenario is expanded in detail in Section 6.

ID	Scenario Name	Description
SC_A01	Administrator logs in	Administrator logs in using username and password and is redirected to the admin dashboard.
SC_A02	Register a new patient	Administrator registers a new patient by filling out the registration form with personal, contact, and insurance details.
SC_A03	Search for a patient	Administrator searches the patient directory by name, ID, or treatment type to locate a specific record.
SC_A04	Update patient information	Administrator updates a patient's personal or insurance details in their profile.
SC_A05	Add a new doctor	Administrator adds a new doctor to the system with their profile and weekly availability schedule.
SC_A06	Deactivate a doctor	Administrator deactivates a doctor who has left the clinic, reassigning their future appointments.
SC_A07	Create an appointment	Administrator creates an appointment on behalf of a patient by selecting doctor, date, time, and service.
SC_A08	Reschedule an appointment	Administrator reschedules an existing appointment to a new date/time and notifies patient and doctor.
SC_A09	Cancel an appointment	Administrator cancels an appointment, releases the room, and notifies patient and doctor.
SC_A10	Assign a room to an appointment	Administrator assigns an available consultation or procedure room to a confirmed appointment.
SC_A11	Add/update service catalog	Administrator adds a new clinic service or updates the price/description of an existing service.
SC_A12	Generate a patient bill	Administrator generates an itemized bill for a patient covering all unbilled services.
SC_A13	Record a payment	Administrator records a payment (cash, card, transfer, insurance) against a patient's pending bill.
SC_A14	Generate a report	Administrator generates an occupancy, revenue, or service utilization report for a selected date range.
SC_A15	View admin dashboard	Administrator views the daily overview dashboard with appointments, room availability, pending bills, and recent registrations.

6. User Scenarios Extended

Each user scenario is expanded below with step-by-step flows, including both the normal path and exception handling.

SC_A01: Administrator logs in

1. Administrator navigates to the login page.
2. Administrator enters username and password.
3. Administrator clicks 'Login'.
4. System validates credentials.
5. System identifies the user role as Administrator.
6. System redirects to the Administrator dashboard.
7. If credentials are invalid: system displays an error message and prompts retry.
8. After 5 failed attempts: system locks the account and displays a contact-support message.

SC_A02: Register a new patient

1. Administrator clicks 'Register New Patient' from the Patient Management section.
2. System displays the registration form.
3. Administrator enters personal details: full name, date of birth, gender, national ID number.
4. Administrator enters contact details: phone, email, address.
5. Administrator enters insurance info or marks as 'Self-Pay'.
6. Administrator enters known allergies and brief medical history notes.
7. Administrator clicks 'Submit Registration'.
8. System validates all required fields.
9. System checks for duplicate patients (matching name + DOB or national ID).
10. If duplicate found: system warns and shows existing record; administrator confirms or cancels.
11. System creates the patient record with a unique Patient ID.
12. System sends welcome notification with portal credentials to the patient.
13. System displays confirmation with the new Patient ID.

SC_A03: Search for a patient

1. Administrator navigates to the Patient List.
2. Administrator types a search term (name, ID, or treatment type) into the search bar.
3. System returns matching results dynamically as the administrator types (after 2+ characters).
4. Administrator can apply filters (e.g., IVF patients, appointments this week).
5. Results show patient name, ID, contact, and next appointment.
6. Administrator clicks a result to open the full patient profile.

SC_A04: Update patient information

1. Administrator opens a patient's profile (via search or patient list).
2. Administrator clicks 'Edit Profile'.
3. Administrator updates the relevant fields (contact, address, insurance).
4. Administrator clicks 'Save Changes'.

5. System validates the updated fields.
6. System saves the changes with a timestamp and administrator's identity.
7. Previous values are stored in the audit history.
8. Updated insurance details apply to future bill calculations.

SC_A05: Add a new doctor

1. Administrator navigates to 'Staff Management' > 'Doctors'.
2. Administrator clicks 'Add New Doctor'.
3. System displays the doctor registration form.
4. Administrator enters: full name, specialization, license number, contact details, qualifications.
5. Administrator configures the weekly availability schedule (days and time slots).
6. Administrator clicks 'Save'.
7. System validates data and checks license number uniqueness.
8. If license number exists: system shows error; administrator corrects.
9. System creates the doctor profile and generates login credentials.
10. Doctor appears in the appointment booking list.
11. System sends credentials to the doctor via email.

SC_A06: Deactivate a doctor

1. Administrator navigates to 'Staff Management' > 'Doctors'.
2. Administrator selects the doctor to deactivate.
3. Administrator clicks 'Deactivate'.
4. System checks for future appointments assigned to this doctor.
5. If future appointments exist: system displays them and requires reassignment.
6. Administrator reassigns each appointment to another doctor or cancels them.
7. System marks the doctor as 'Inactive'.
8. Doctor is removed from the booking list but remains in historical records.

SC_A07: Create an appointment

1. Administrator navigates to 'Appointment Management'.
2. Administrator clicks an available time slot on the calendar (or clicks 'New Appointment').
3. System displays the appointment creation form.
4. Administrator searches for and selects the patient.
5. Administrator selects the doctor.
6. Administrator selects the service type.
7. Administrator adds optional notes (reason for visit).
8. System checks for scheduling conflicts.
9. If conflict found: system suggests alternative slots.
10. System creates the appointment with status 'Confirmed'.
11. System sends notifications to both patient and doctor.

SC_A08: Reschedule an appointment

1. Administrator clicks on an existing appointment in the calendar.
2. Administrator clicks 'Reschedule'.
3. Administrator selects a new date and time.
4. System checks for conflicts.
5. If conflict: system suggests alternatives.
6. System updates the appointment.
7. If a room was assigned: system reassigns the room or releases it.
8. System notifies the patient and doctor of the change.

SC_A09: Cancel an appointment

1. Administrator clicks on an existing appointment.
2. Administrator clicks 'Cancel'.
3. System prompts for a cancellation reason (mandatory).
4. Administrator enters the reason and confirms.
5. System cancels the appointment and releases the time slot.
6. If a room was assigned: system releases the room.
7. System notifies the patient and doctor.

SC_A10: Assign a room to an appointment

1. Administrator selects a confirmed appointment (from Appointments or Room Management).
2. Administrator clicks 'Assign Room'.
3. System displays rooms available for the appointment's time window, filtered by type.
4. Administrator selects a room.
5. System checks for conflicts.
6. If conflict (room just taken): system refreshes the list; administrator picks another.
7. System assigns the room and updates its status to 'Occupied'.
8. Room assignment is visible to doctor and patient.

SC_A11: Add/update service catalog

Adding a service:

1. Administrator navigates to 'Services'.
2. Administrator clicks 'Add New Service'.
3. Administrator enters: name, category, description, price.
4. Administrator clicks 'Save'.
5. System validates (checks for duplicate name in same category).
6. Service is added with status 'Active' and available for billing.

Updating a service:

1. Administrator selects a service.
2. Administrator edits price, description, or category.
3. Administrator clicks 'Save Changes'.
4. New price applies to future bills; existing invoices are unchanged.

Deactivating a service:

1. Administrator selects a service and clicks 'Deactivate'.
2. System marks it as 'Inactive'; it no longer appears in billing options.
3. Historical bills are preserved.

SC_A12: Generate a patient bill

1. Administrator navigates to 'Billing'.
2. Administrator searches for and selects the patient.
3. System retrieves all unbilled services.
4. System displays the itemized list with prices, quantities, and line totals.
5. System calculates subtotal, taxes, and insurance deductions (if applicable).
6. Administrator reviews each line item.
7. Administrator optionally applies a discount (reason required).
8. Administrator clicks 'Generate Bill'.
9. System creates the invoice with status 'Pending' and assigns an invoice number.
10. Bill is visible in the patient's portal.
11. System sends a billing notification to the patient.

SC_A13: Record a payment

1. Administrator navigates to 'Billing' > 'Payments'.
2. Administrator searches for the patient.
3. System displays outstanding bills.
4. Administrator selects the bill.
5. System shows bill details and remaining balance.
6. Administrator enters payment amount and selects method (cash, card, transfer, insurance).
7. Administrator enters reference number if applicable.
8. Administrator clicks 'Record Payment'.
9. System validates (amount cannot exceed balance).
10. System records the transaction.
11. System updates bill status ('Paid' or 'Partially Paid').
12. System generates a receipt available to the patient.

SC_A14: Generate a report

1. Administrator navigates to 'Reports'.
2. Administrator selects report type (Occupancy, Revenue, or Service Utilization).
3. Administrator enters the date range.
4. Administrator optionally adds filters (department, doctor, service category).
5. Administrator clicks 'Generate Report'.
6. System processes data and displays the report with summary metrics and tables.
7. Administrator can export the report as PDF.
8. System saves the report in report history.

SC_A15: View admin dashboard

1. Administrator logs in (redirected to dashboard) or clicks 'Dashboard' in navigation.
2. System loads the dashboard within 2 seconds.
3. Dashboard shows: today's appointment count, checked-in patients, available rooms, pending bill count and total outstanding amount, recent registrations (last 7 days).
4. Administrator can click any section to navigate to its detailed view.

7. Use Cases – Administrator User Flows

Each use case describes the full user flow for the administrator, following the structured format from the course case study. Each includes: description, preconditions, main flow, alternative/exception paths, what can go wrong, postconditions, restrictions and constraints, and related user stories.

7.1 UC_A01: Register New Patient

UC Name	Register New Patient
UC ID	UC_A01
Actors	Hospital Administrator
Description	The administrator registers a new patient in the clinic system by entering all required personal, contact, and insurance information. The system creates a patient record that enables the patient to book appointments, receive treatment, and access their portal.
Preconditions	Administrator is authenticated and logged in. The patient is physically present or has provided their details remotely. The patient does not already have an active record in the system.
Main Flow (Normal)	1. Administrator navigates to 'Patient Management' from the dashboard. 2. Administrator clicks the 'Register New Patient' button. 3. System displays the patient registration form. 4. Administrator enters personal details: full name, date of birth, gender, national ID number. 5. Administrator enters contact details: phone number, email, address. 6. Administrator enters insurance information: provider name, policy number, coverage type (or marks as 'Self-Pay'). 7. Administrator enters brief medical history notes and known allergies. 8. Administrator clicks 'Submit Registration'. 9. System validates all required fields and checks for duplicates. 10. System creates the patient record and generates a unique Patient ID. 11. System generates portal login credentials for the patient. 12. System sends a welcome notification (email/SMS) to the patient with their Patient ID and login details. 13. System displays a confirmation screen with the new Patient ID.
Alternative / Exception Flow	9a. Validation fails: - System highlights the invalid/missing fields with error messages. - Administrator corrects the data and resubmits. 9b. Duplicate detected: - System displays a warning showing the existing patient's name, DOB, and ID. - Administrator can choose to open the existing record or confirm this is a new patient. 6a. Insurance details unavailable: - Administrator marks billing type as 'Self-Pay'.

	- Insurance fields become optional.
What Can Go Wrong	The administrator may enter incorrect patient data (e.g., wrong DOB or ID number), leading to billing or identity issues later. Mitigation: mandatory field validation and duplicate detection. Network failure during submission could lose entered data. Mitigation: form data is auto-saved locally every 30 seconds. The patient's email/phone may be invalid, causing notification delivery failure. Mitigation: system logs notification failures for administrator follow-up.
Postconditions	A new patient record exists with a unique Patient ID. The patient can log in to the portal. The patient can now book appointments and receive services. The registration event is logged in the audit trail.
Restrictions & Constraints	National ID number must be unique across all patient records. All mandatory fields (full name, DOB, contact number, national ID) must be completed. Patient data must comply with local data protection laws. Passwords for the patient portal must be stored as hashed values. Only users with the Administrator role can register patients.
Related User Stories	US_A01, US_A10, US_A11
Dependencies	None — this is an entry-point use case. A patient record does not need any other entity to exist before registration.

7.2 UC_A02: Assign Room to Appointment

UC Name	Assign Room to Appointment
UC ID	UC_A02
Actors	Hospital Administrator
Description	The administrator assigns a specific consultation or procedure room to a patient's confirmed appointment so that the facility is organized and no two appointments occupy the same room simultaneously.
Preconditions	Administrator is authenticated and logged in. The appointment exists with status 'Confirmed'. At least one room of the appropriate type is available for the appointment time slot.
Main Flow (Normal)	1. Administrator navigates to 'Appointments' or 'Room Management'. 2. Administrator selects the appointment that needs a room assignment. 3. System displays the appointment details (patient name, doctor, date, time, service type). 4. Administrator clicks 'Assign Room'.

	5. System retrieves and displays rooms available for the appointment's time window, filtered by room type (consultation, procedure, lab). 6. Administrator selects a room from the available list. 7. System checks for scheduling conflicts with the selected room. 8. System assigns the room to the appointment. 9. System updates the room status to 'Occupied' for the appointment's time window. 10. System displays confirmation with room number and appointment details. 11. Doctor and patient can see the assigned room in their respective views.
Alternative / Exception Flow	5a. No rooms available: - System displays a message indicating all rooms are occupied for the time window. - System suggests the nearest available time slots for each room. - Administrator can choose to reschedule the appointment. 7a. Conflict detected (race condition): - System notifies the administrator that the room was just assigned to another appointment. - System refreshes the available room list. - Administrator selects a different room.
What Can Go Wrong	Two administrators may try to assign the same room simultaneously. Mitigation: optimistic locking with conflict detection at save time. A room may have been marked as available but is under maintenance. Mitigation: maintenance status overrides availability. The appointment may be cancelled after room assignment. Mitigation: cancellation workflow automatically releases the room.
Postconditions	The appointment has an assigned room. The room's schedule reflects the booking. Doctor and patient are informed of the room assignment. The assignment is logged.
Restrictions & Constraints	A room cannot be assigned to overlapping time windows. Only rooms matching the appointment type (consultation vs. procedure) can be assigned. Rooms under maintenance cannot be assigned. Only administrators can assign rooms. Room assignment changes must notify both doctor and patient.
Related User Stories	US_A02, US_A07, US_A12
Dependencies	UC_A01 (Register New Patient) — a patient must exist. UC_A04 (Manage Doctors) — a doctor must exist. UC_A06 (Manage Appointment Schedules) — a confirmed appointment must exist before a room can be assigned to it.

7.3 UC_A03: Generate Patient Bill

UC Name	Generate Patient Bill
UC ID	UC_A03
Actors	Hospital Administrator
Description	The administrator generates an itemized bill for a patient covering all services rendered during their visit or treatment cycle.
Preconditions	Administrator is authenticated and logged in. The patient exists in the system. The patient has at least one unbilled service recorded. Service prices are configured in the service catalog.
Main Flow (Normal)	1. Administrator navigates to 'Billing' from the dashboard. 2. Administrator searches for and selects the patient. 3. System retrieves all unbilled services associated with the patient. 4. System displays the itemized list with: service name, date of service, unit price, quantity, and line total. 5. System calculates the subtotal. 6. System applies applicable taxes. 7. If patient has insurance: system calculates the covered amount and shows the patient's co-pay. 8. System displays the total amount due. 9. Administrator reviews each line item for accuracy. 10. Administrator optionally applies a discount (must enter a reason). 11. Administrator clicks 'Generate Bill'. 12. System creates the invoice with status 'Pending'. 13. System assigns an invoice number. 14. System makes the bill visible in the patient's portal. 15. System sends a billing notification to the patient (email/SMS).
Alternative / Exception Flow	3a. No unbilled services: - System displays: 'No pending charges found for this patient.' 9a. Price discrepancy found: - Administrator flags a line item. - Administrator can override the price with a mandatory reason (logged for audit). 7a. Insurance information missing or expired: - System warns the administrator. - Administrator can proceed with full patient responsibility or update insurance details first. Partial billing: Administrator can select specific services to bill now, leaving others for later.
What Can Go Wrong	A service may have been recorded with an incorrect price. Mitigation: price override with mandatory reason and audit logging. Insurance calculation may be incorrect due to outdated policy data. Mitigation: administrator can manually adjust the covered amount. The patient may dispute charges. Mitigation: bills cannot be deleted, only credited/adjusted.

Postconditions	An itemized invoice with a unique number is created. The invoice is linked to the patient's account. The patient can view the bill in their portal. Financial records are updated. All line items, discounts, and overrides are logged.
Restrictions & Constraints	Only administrators can generate and modify bills. All price overrides require a written reason and are permanently logged. Bills cannot be deleted; corrections are made via credit notes. Billing must comply with local tax regulations. Generated bills must be retained for a minimum of 5 years.
Related User Stories	US_A03, US_A06, US_A08, US_A09
Dependencies	UC_A01 (Register New Patient) — the patient must be registered. UC_A05 (Manage Service Catalog and Prices) — services and their prices must be configured before they can be billed. UC_A06 (Manage Appointment Schedules) — services are typically rendered through appointments that must already exist.

7.4 UC_A04: Manage Doctors

UC Name	Manage Doctors
UC ID	UC_A04
Actors	Hospital Administrator
Description	The administrator adds new doctors, updates existing doctor profiles, or deactivates doctors who are no longer practicing at the clinic.
Preconditions	Administrator is authenticated and logged in. For adding: the doctor's professional details and license number are available. For editing/deactivating: the doctor's profile exists in the system.
Main Flow (Normal)	Adding a new doctor: 1. Administrator navigates to 'Staff Management' > 'Doctors'. 2. Administrator clicks 'Add New Doctor'. 3. System displays the doctor registration form. 4. Administrator enters: full name, specialization, license number, contact details, qualifications. 5. Administrator configures the weekly availability schedule. 6. Administrator clicks 'Save'. 7. System validates the data (checks license number uniqueness). 8. System creates the doctor profile. 9. Doctor appears in the appointment booking list. 10. System generates login credentials and sends them to the doctor.

	Editing an existing doctor: 1. Administrator selects a doctor from the list. 2. Administrator updates the necessary fields. 3. Administrator clicks 'Save Changes'. 4. System validates and saves the updates. Deactivating a doctor: 1. Administrator selects a doctor and clicks 'Deactivate'. 2. System checks for future appointments. 3. If future appointments exist: system requires reassignment. 4. Administrator reassigns or cancels each appointment. 5. System marks the doctor as 'Inactive'. 6. Doctor is removed from booking options but stays in historical records.
Alternative / Exception Flow	7a. License number already exists: system displays error; administrator corrects. 3a. (Deactivation) No future appointments: system proceeds directly. Reactivation: administrator can restore a deactivated doctor to active status.
What Can Go Wrong	Deactivating a doctor with many future appointments could disrupt patient care. Mitigation: mandatory reassignment. Incorrect license number could create compliance issues. Mitigation: format validation. Schedule changes may conflict with existing appointments. Mitigation: system warns about conflicts.
Postconditions	Doctor profile is created, updated, or deactivated. Booking list reflects current active doctors. All changes are logged. Affected patients and doctors are notified.
Restrictions & Constraints	License number must be unique. Only administrators can manage doctor profiles. Deactivation cannot proceed until all future appointments are resolved. Deactivated doctors are hidden from booking but retained in historical records.
Related User Stories	US_A04, US_A05, US_A07
Dependencies	None for adding a new doctor — this is an entry-point use case. UC_A06 (Manage Appointment Schedules) — only relevant when deactivating a doctor who has existing future appointments that must be reassigned first.

7.5 UC_A05: Manage Service Catalog and Prices

UC Name	Manage Service Catalog and Prices
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UC ID	UC_A05
Actors	Hospital Administrator
Description	The administrator manages the clinic's catalog of services, including adding new services, updating descriptions and prices, and deactivating obsolete services.
Preconditions	Administrator is authenticated and logged in. The service catalog module is accessible.
Main Flow (Normal)	1. Administrator navigates to 'Services' from the dashboard. 2. System displays the current catalog with name, category, description, price, and status. Adding a new service: 3. Administrator clicks 'Add New Service'. 4. Administrator enters: name, category, description, price. 5. Administrator clicks 'Save'. 6. System validates (checks for duplicate name in same category). 7. Service is added with status 'Active'. Updating a service: 3. Administrator selects a service. 4. Administrator edits price, description, or category. 5. Administrator clicks 'Save Changes'. 6. New price applies to future bills; existing invoices unchanged. Deactivating a service: 3. Administrator selects a service and clicks 'Deactivate'. 4. System marks it as 'Inactive'; no longer available for billing. 5. Historical bills are preserved.
Alternative / Exception Flow	6a. Duplicate name in same category: system shows error. Bulk price update: administrator can select multiple services and apply a percentage change.
What Can Go Wrong	Incorrect price change could cause billing disputes. Mitigation: price history maintained. Deactivating a service still in use disrupts billing. Mitigation: system warns if billed in last 30 days.
Postconditions	Catalog reflects the changes. Active services are available for billing. Inactive services are hidden but preserved in history. All changes are logged.
Restrictions & Constraints	Service name must be unique within its category. Price must be a positive number and is mandatory. Deactivated services require confirmation to reactivate. Price changes do not affect existing invoices.

	Only administrators can modify the catalog.
Related User Stories	US_A06, US_A03, US_A09
Dependencies	None — this is an entry-point use case. The service catalog can be populated independently of any other entity.

7.6 UC_A06: Manage Appointment Schedules

UC Name	Manage Appointment Schedules
UC ID	UC_A06
Actors	Hospital Administrator
Description	The administrator views, creates, modifies, or cancels appointments across all doctors' schedules for centralized control.
Preconditions	Administrator is authenticated and logged in. At least one doctor and one patient exist in the system.
Main Flow (Normal)	1. Administrator navigates to 'Appointment Management'. 2. System displays a calendar view of all doctors' schedules. 3. Administrator filters by doctor, department, or date range. Creating: 4. Administrator clicks an available slot. 5. Administrator selects patient, doctor, service type, adds notes. 6. System checks conflicts. 7. System creates appointment ('Confirmed') and notifies patient + doctor. Rescheduling: 4. Administrator clicks existing appointment > 'Reschedule'. 5. Administrator selects new date/time. 6. System checks conflicts, updates, and notifies both parties. Cancelling: 4. Administrator clicks existing appointment > 'Cancel'. 5. Administrator enters reason (mandatory). 6. System cancels, releases room if assigned, and notifies both parties.
Alternative / Exception Flow	Conflict detected: system suggests nearest available slots. Patient not found: system suggests registering first (UC_A01). Walk-in: administrator creates same-day appointment marked as 'Walk-in'.
What Can Go Wrong	Cancellation without notification. Mitigation: automatic notifications. Overbooking via concurrent access. Mitigation: optimistic concurrency control.

Postconditions	Late rescheduling not reaching patient. Mitigation: warning for changes within 2 hours.
	Appointment is created, rescheduled, or cancelled. Schedules are updated; rooms adjusted. Both parties are notified. All changes are logged.
Restrictions & Constraints	Appointments only during doctor's configured hours. No overlapping bookings per doctor. Cancellation requires a reason. Notifications within 1 minute of any change. Only administrators and patients (for their own) can modify schedules.
Related User Stories	US_A07, US_A02, US_A12
Dependencies	UC_A01 (Register New Patient) — a patient must exist. UC_A04 (Manage Doctors) — a doctor must exist with a configured availability schedule. UC_A05 (Manage Service Catalog and Prices) — the service type selected for the appointment must exist in the catalog.

7.7 UC_A07: Record Payment

UC Name	Record Payment
UC ID	UC_A07
Actors	Hospital Administrator
Description	The administrator records a payment made by a patient against a pending or partially paid bill.
Preconditions	Administrator is authenticated and logged in. Patient has at least one bill with status 'Pending' or 'Partially Paid'. Payment has been received.
Main Flow (Normal)	1. Administrator navigates to 'Billing' > 'Payments'. 2. Administrator searches for the patient. 3. System displays outstanding bills. 4. Administrator selects the bill. 5. System displays bill details and remaining balance. 6. Administrator enters payment amount. 7. Administrator selects payment method (cash, card, bank transfer, insurance). 8. Administrator enters reference number if applicable. 9. Administrator clicks 'Record Payment'. 10. System validates payment amount (cannot exceed balance). 11. System records the transaction. 12. System updates bill status ('Paid' or 'Partially Paid').

	13. System generates a receipt. 14. Receipt is available in the patient's portal. 15. System displays confirmation with updated balance.
Alternative / Exception Flow	10a. Amount exceeds balance: system shows error. 3a. No outstanding bills: system displays message. Refund: administrator can issue a refund with mandatory reason. Insurance: administrator records insurance reimbursement separately from co-pay.
What Can Go Wrong	Wrong amount recorded. Mitigation: validation and confirmation step. Bank transfer not cleared. Mitigation: 'Pending Verification' status option. Payment against wrong bill. Mitigation: system shows bill details for verification.
Postconditions	Payment recorded with amount, method, date, and reference. Bill status updated. Receipt generated. Patient's financial summary updated. Transaction logged.
Restrictions & Constraints	Amount cannot exceed outstanding balance. Payment method is mandatory. Refunds require a reason. Records cannot be deleted; corrections via refund transactions. Financial records retained for 5+ years. Only administrators can record payments.
Related User Stories	US_A08, US_A03, US_A09
Dependencies	UC_A01 (Register New Patient) — the patient must be registered. UC_A03 (Generate Patient Bill) — a bill with status 'Pending' or 'Partially Paid' must exist before a payment can be recorded against it.

7.8 UC_A08: Generate Reports

UC Name	Generate Reports
UC ID	UC_A08
Actors	Hospital Administrator
Description	The administrator generates occupancy, revenue, or service utilization reports for a selected date range.
Preconditions	Administrator is authenticated and logged in. Transactional data exists for the selected date range.
Main Flow (Normal)	1. Administrator navigates to 'Reports'. 2. System displays report types: Occupancy, Revenue, Service Utilization. 3. Administrator selects type.

	4. Administrator enters date range. 5. Administrator optionally adds filters. 6. Administrator clicks 'Generate Report'. 7. System processes data. 8. System displays report with metrics and tables. 9. Administrator can export as PDF. 10. System saves the report in history.
Alternative / Exception Flow	7a. No data for range: system displays message; administrator adjusts. Comparison: administrator selects two date ranges for side-by-side analysis.
What Can Go Wrong	Incomplete data may cause inaccurate reports. Mitigation: system flags data gaps. Large ranges slow generation. Mitigation: progress indicator; async for 1+ year ranges. PDF formatting issues. Mitigation: standardized template.
Postconditions	Report generated and displayed. Saved in report history. PDF available if exported. No data modified.
Restrictions & Constraints	Reports are read-only. Date range max 2 years. Only administrators and managers can generate. Generation within 10 seconds for up to 3 months. Exported PDFs include generation date and administrator name.
Related User Stories	US_A09, US_A03, US_A08
Dependencies	UC_A03 (Generate Patient Bill) — billing data is required for revenue reports. UC_A06 (Manage Appointment Schedules) — appointment data is required for occupancy reports. UC_A07 (Record Payment) — payment data is required for revenue reports.

8. Use Case Diagram

The Use Case Diagram below describes the Administrator actor and all use cases they interact with. The diagram follows UML notation with «include» and «extend» relationships where applicable.

Actor:

- Hospital Administrator (Armend)

Primary Use Cases (direct interactions):

- Register New Patient – Administrator creates a new patient record in the system.
- Search / Filter Patients – Administrator locates patient records.
- Update Patient Information – Administrator edits patient personal/insurance data.
- Add New Doctor – Administrator registers a new doctor with availability schedule.
- Update / Deactivate Doctor – Administrator edits or deactivates a doctor profile.
- Create Appointment – Administrator books an appointment on behalf of a patient.
- Reschedule Appointment – Administrator changes the date/time of an existing appointment.
- Cancel Appointment – Administrator cancels an appointment with a mandatory reason.
- Assign Room – Administrator assigns a room to a confirmed appointment.
- Manage Service Catalog – Administrator adds, edits, or deactivates clinic services and prices.
- Generate Patient Bill – Administrator creates an itemized invoice for a patient.
- Record Payment – Administrator records a payment against a patient's bill.
- Generate Reports – Administrator creates occupancy, revenue, or utilization reports.
- View Dashboard – Administrator views the daily operations overview.

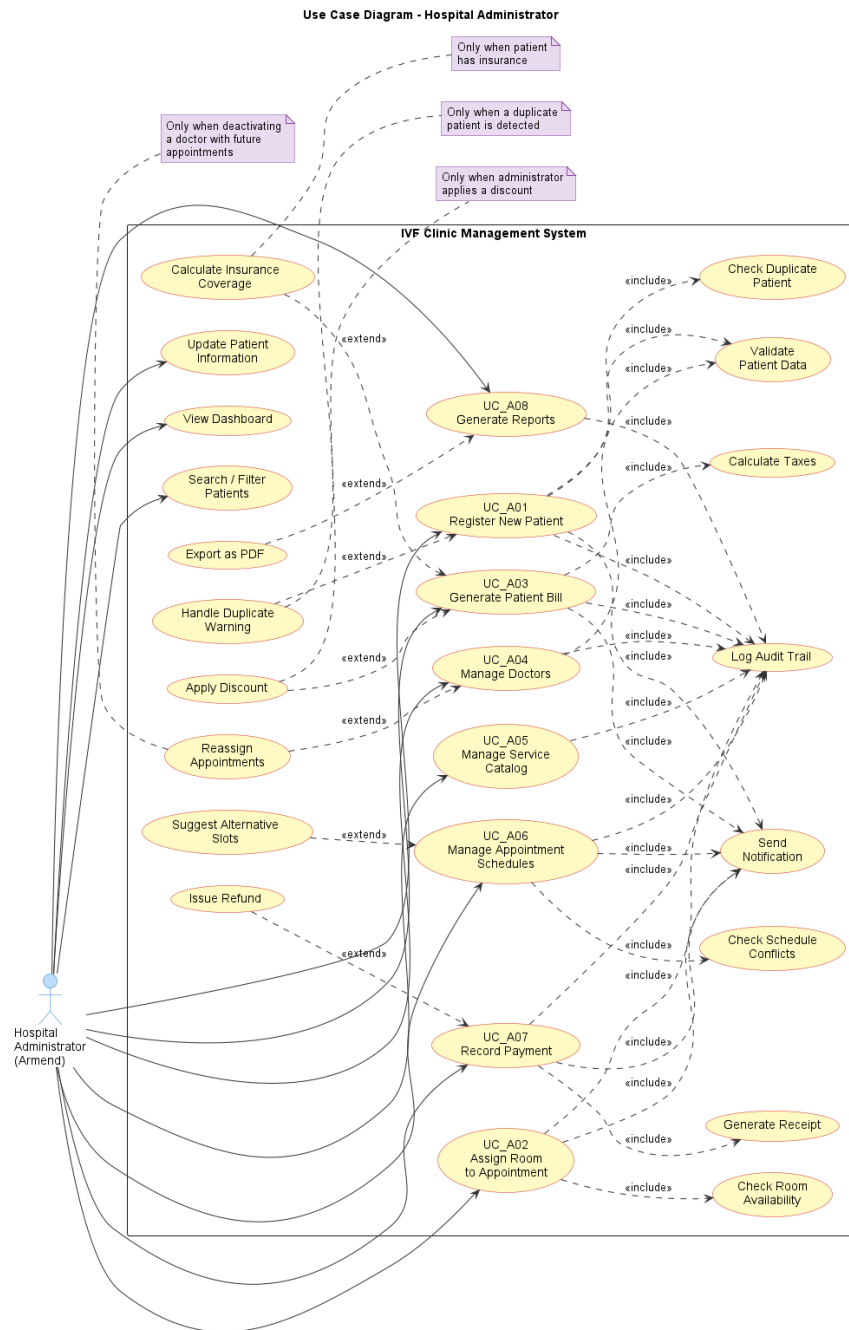
«include» Relationships:

- Register New Patient «include» Validate Patient Data – Patient data is always validated during registration.
- Register New Patient «include» Check Duplicate Patient – Duplicate detection runs automatically.
- Register New Patient «include» Send Notification – Welcome notification is always sent after registration.
- Create Appointment «include» Check Schedule Conflicts – Conflict detection runs for every appointment creation.
- Create Appointment «include» Send Notification – Both patient and doctor are notified.
- Generate Patient Bill «include» Calculate Taxes – Tax calculation is mandatory for every bill.
- Generate Patient Bill «include» Send Notification – Billing notification is sent to the patient.
- Record Payment «include» Generate Receipt – A receipt is always generated after payment.
- Assign Room «include» Check Room Availability – Availability check runs for every assignment.

«extend» Relationships:

- Register New Patient «extend» Handle Duplicate Warning – Extends only when a duplicate is detected.
- Generate Patient Bill «extend» Apply Discount – Extends only when administrator applies a discount.

- Generate Patient Bill «extend» Calculate Insurance Coverage – Extends only when patient has insurance.
- Create Appointment «extend» Suggest Alternative Slots – Extends only when a conflict is detected.
- Deactivate Doctor «extend» Reassign Appointments – Extends only when the doctor has future bookings.
- Record Payment «extend» Issue Refund – Extends only when a refund is requested.
- Generate Reports «extend» Export as PDF – Extends only when administrator requests export.

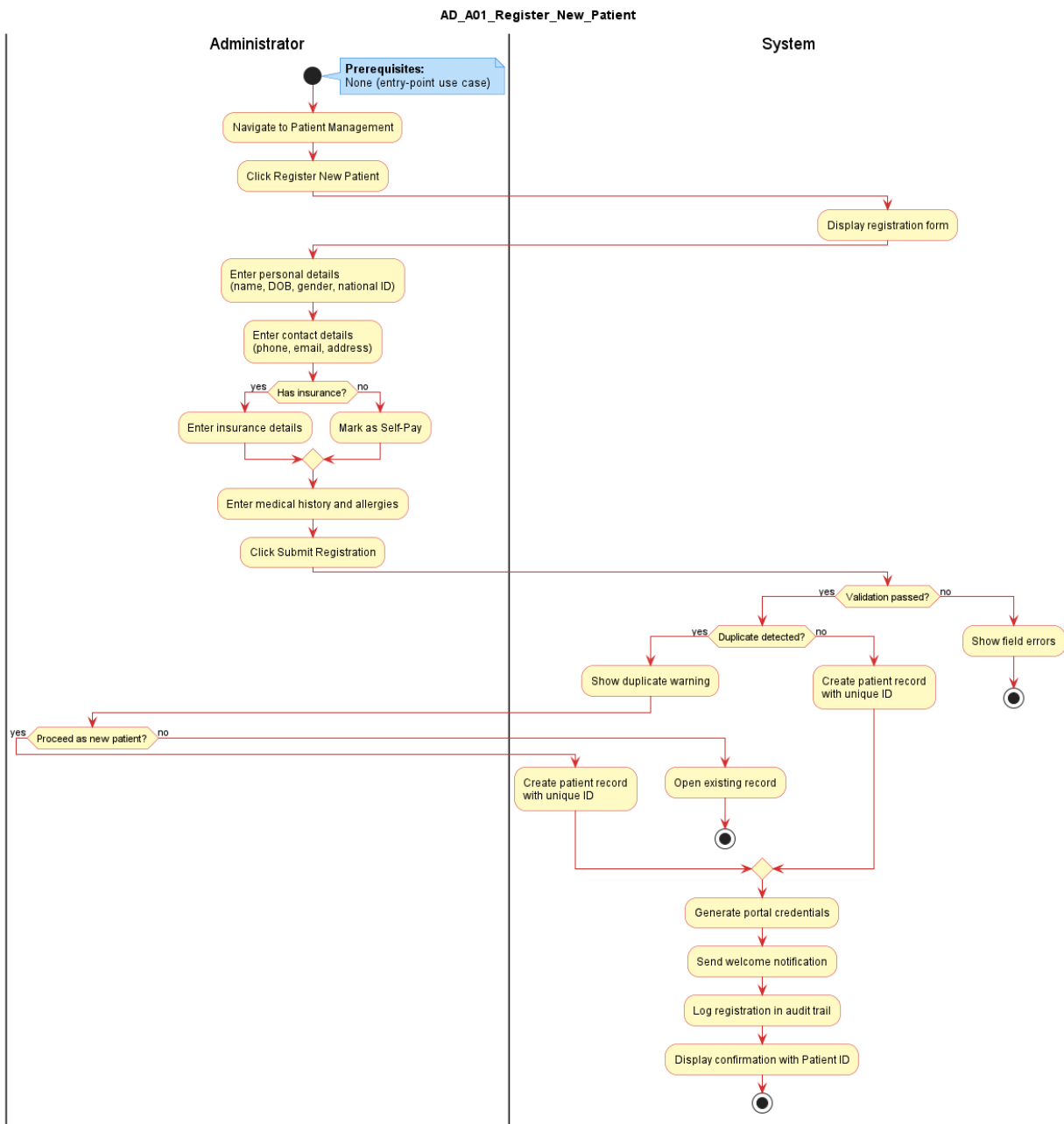


9. Activity Diagrams

This section contains the activity diagrams for each administrator use case. Each diagram illustrates the workflow of the use case including decision points, alternative paths, and exception handling. Insert the corresponding PNG image in the placeholder below each title.

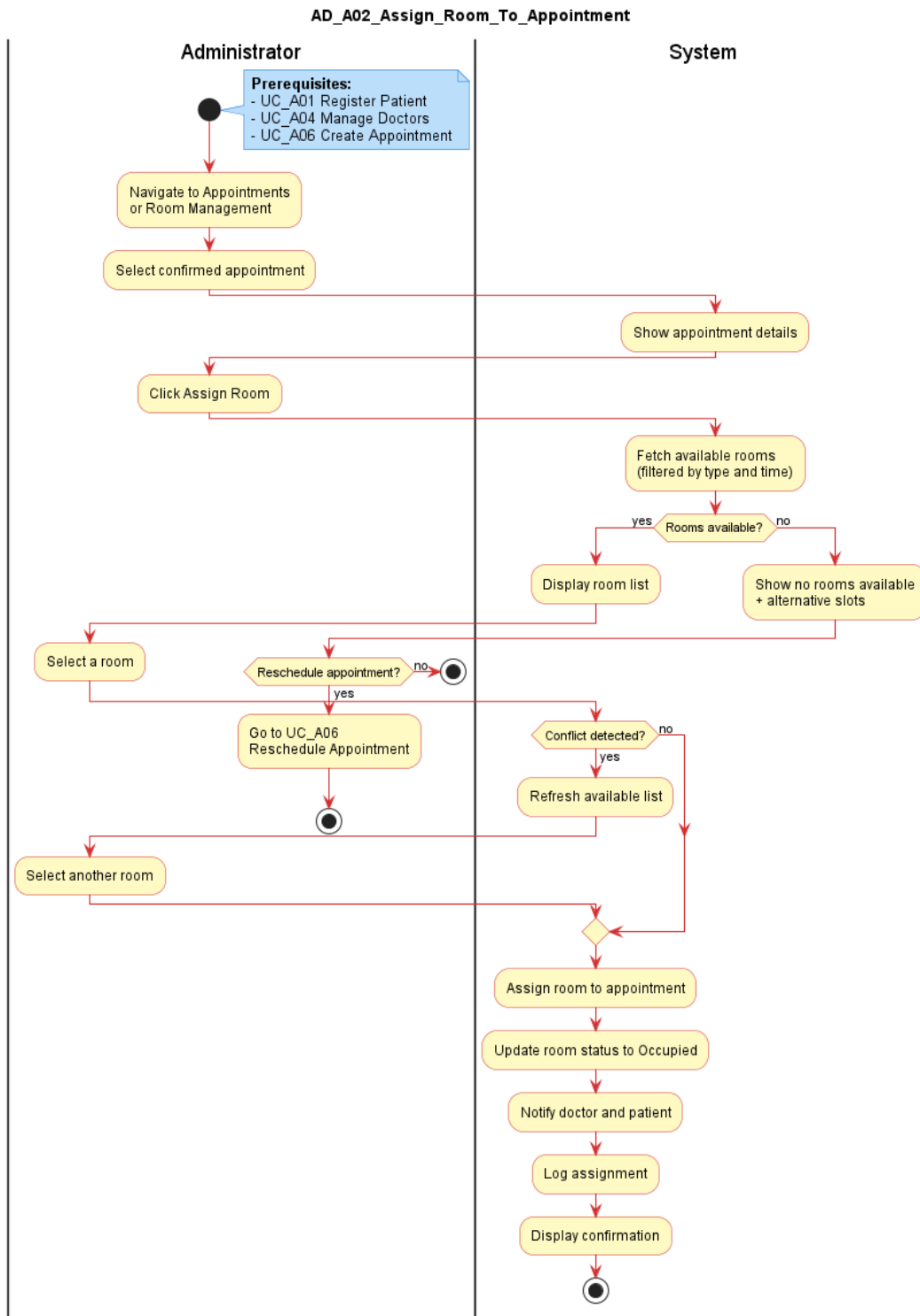
9.1 UC_A01: Register New Patient

Activity diagram for UC_A01 – Register New Patient



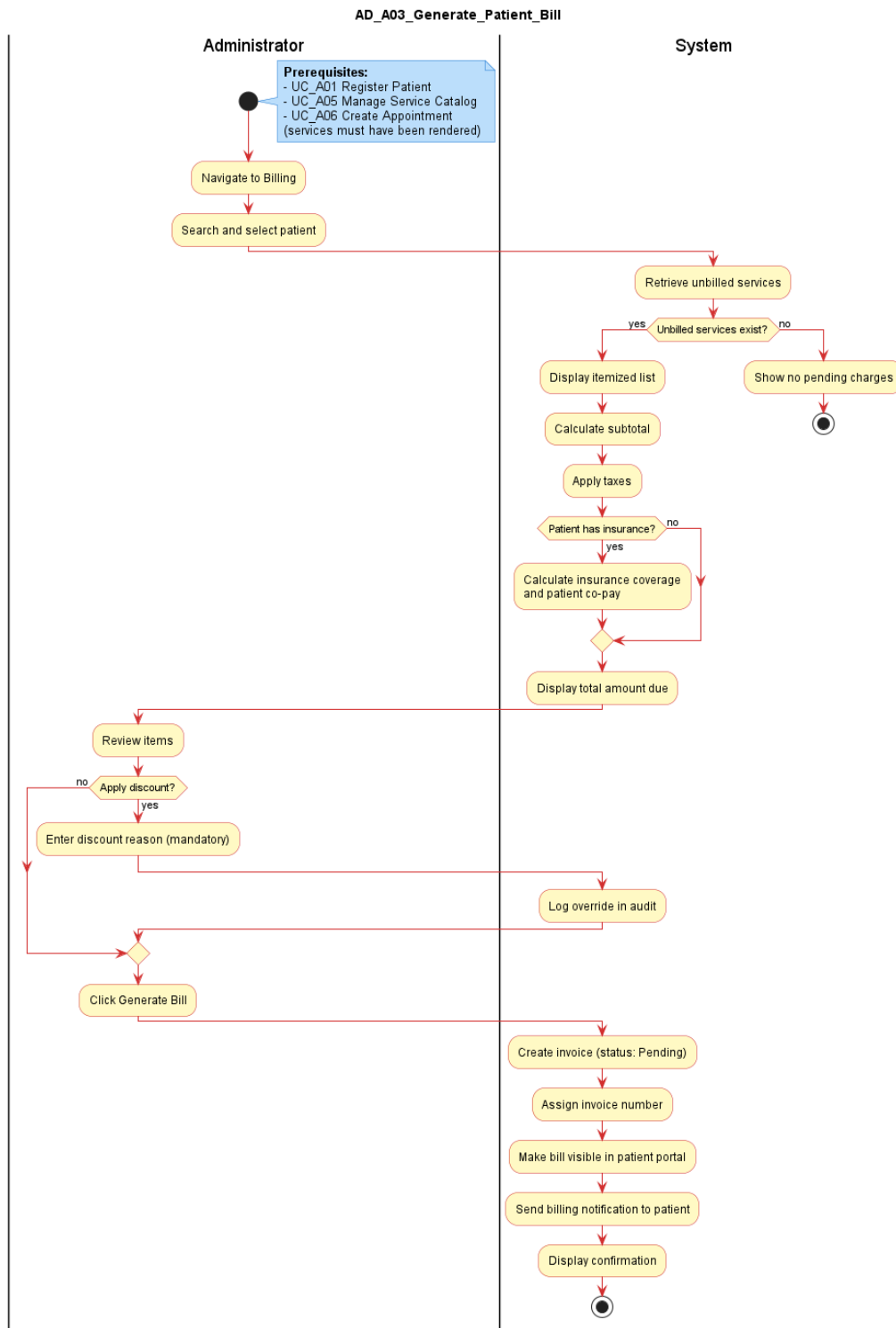
9.2 UC_A02: Assign Room to Appointment

Activity diagram for UC_A02 – Assign Room to Appointment



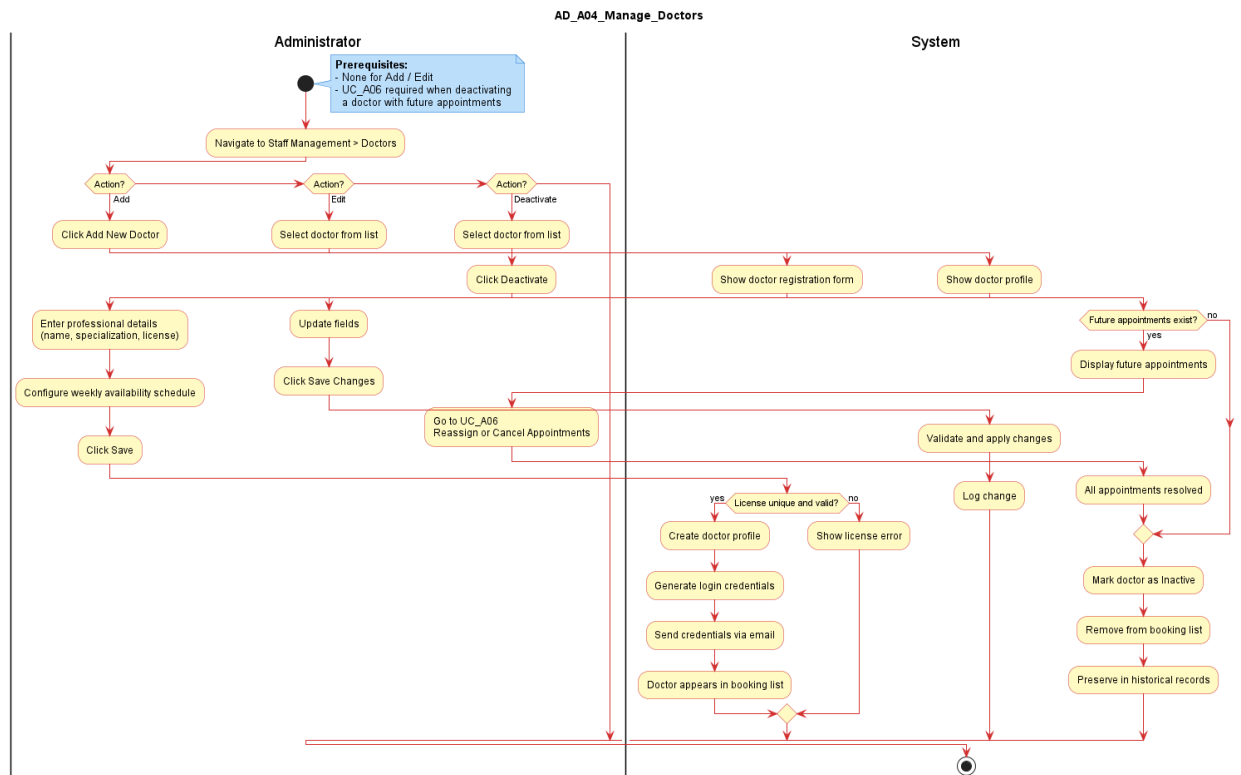
9.3 UC_A03: Generate Patient Bill

Activity diagram for UC_A03 – Generate Patient Bill



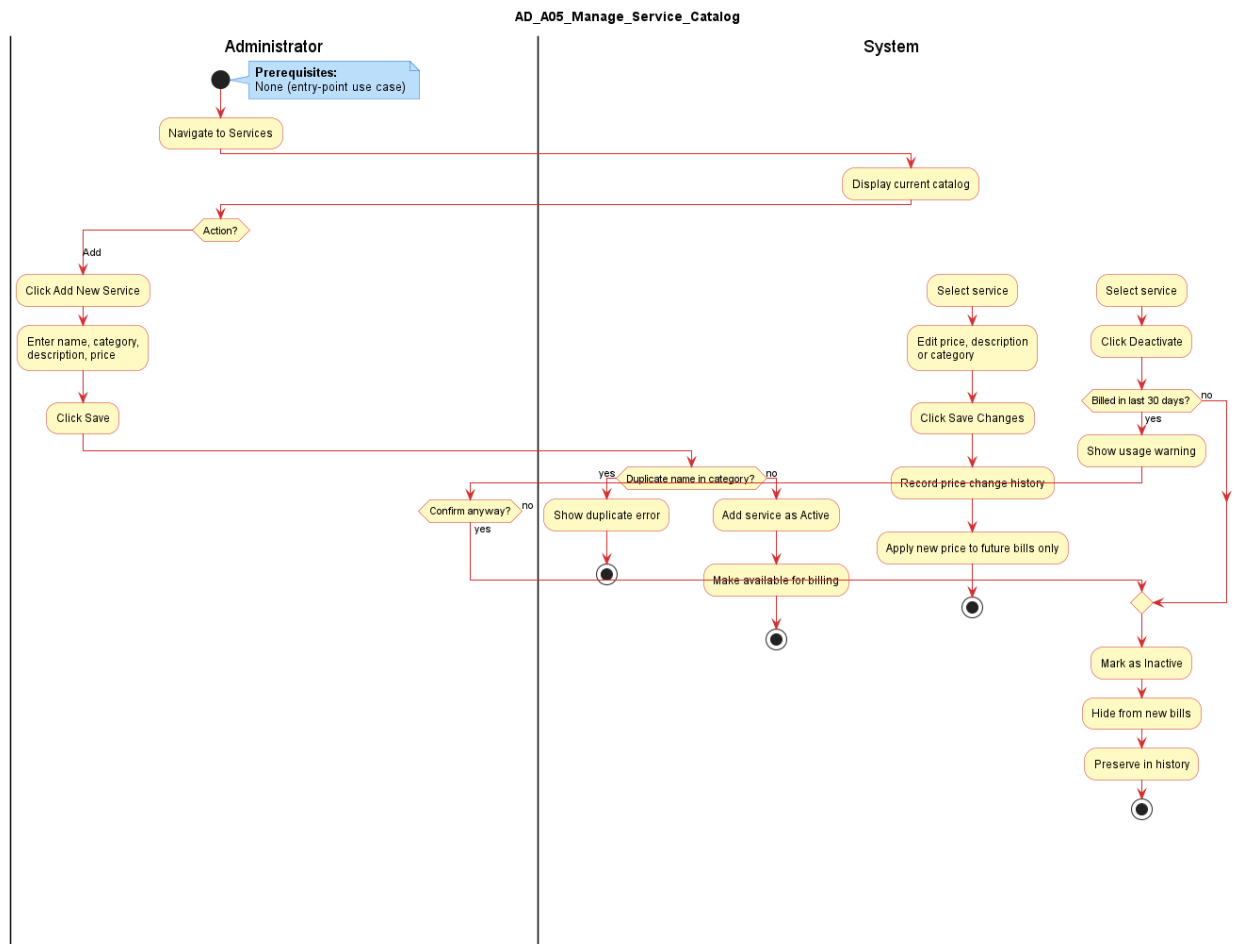
9.4 UC_A04: Manage Doctors (Add / Update / Deactivate)

Activity diagram for UC_A04 – Manage Doctors (Add / Update / Deactivate)



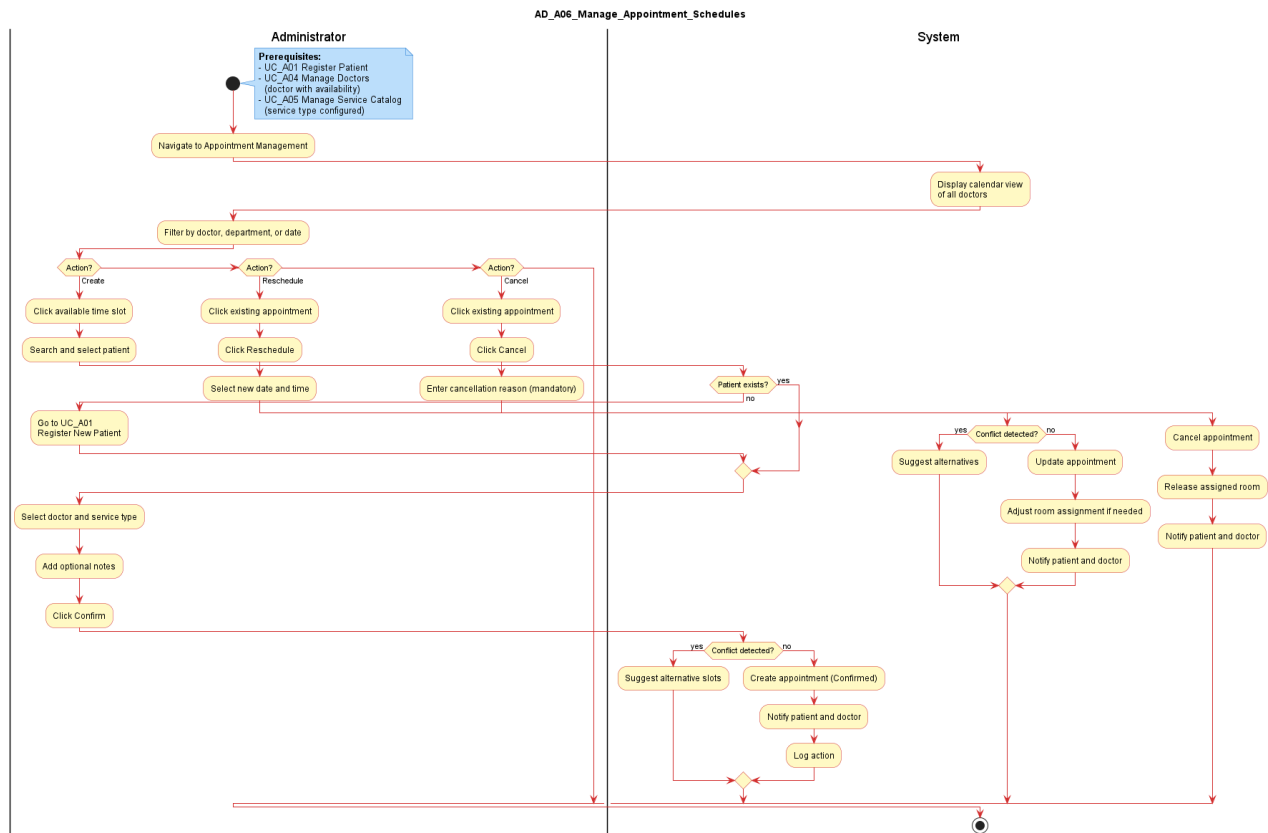
9.5 UC_A05: Manage Service Catalog and Prices

Activity diagram for UC_A05 – Manage Service Catalog and Prices



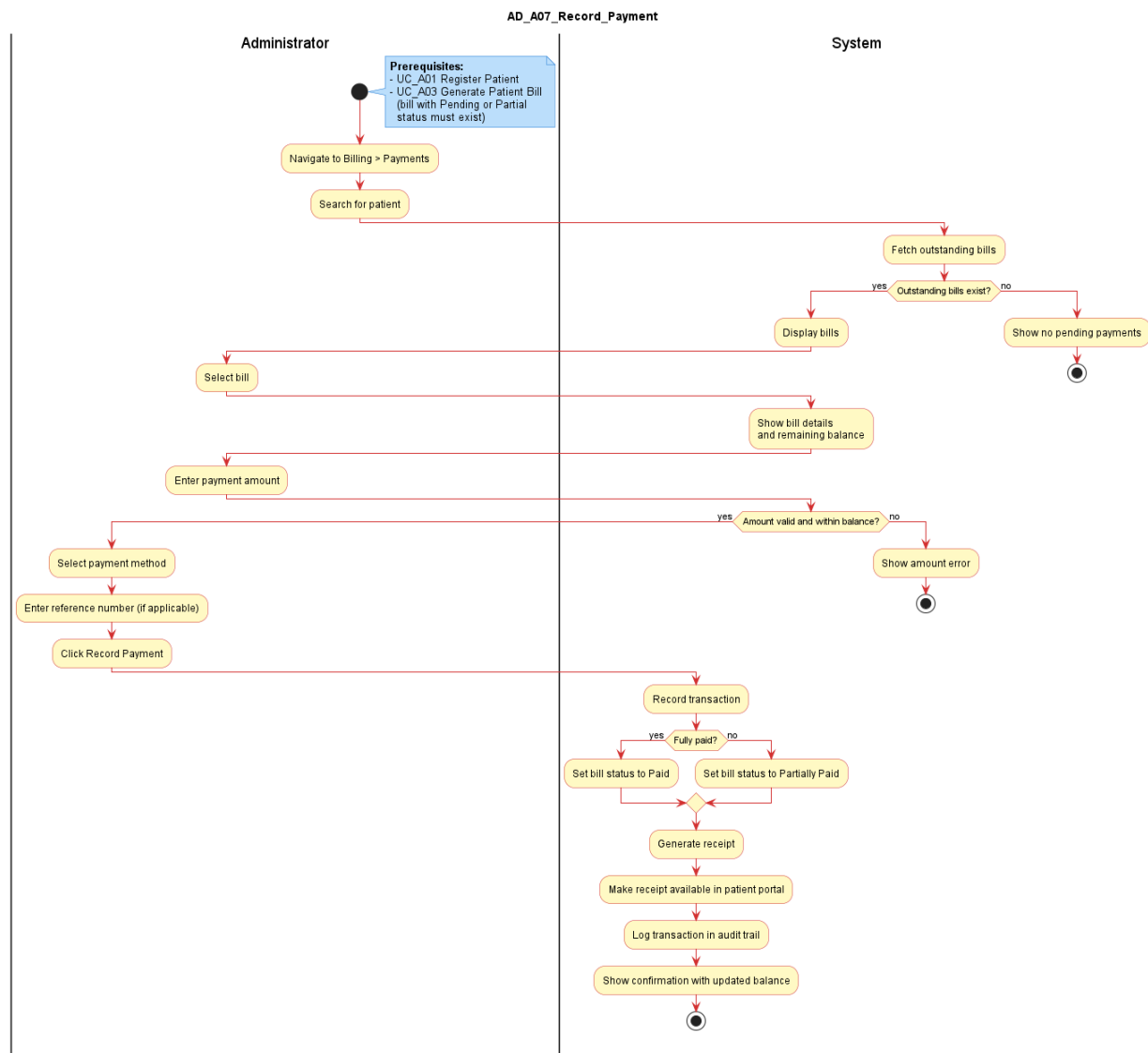
9.6 UC_A06: Manage Appointment Schedules

Activity diagram for UC_A06 – Manage Appointment Schedules



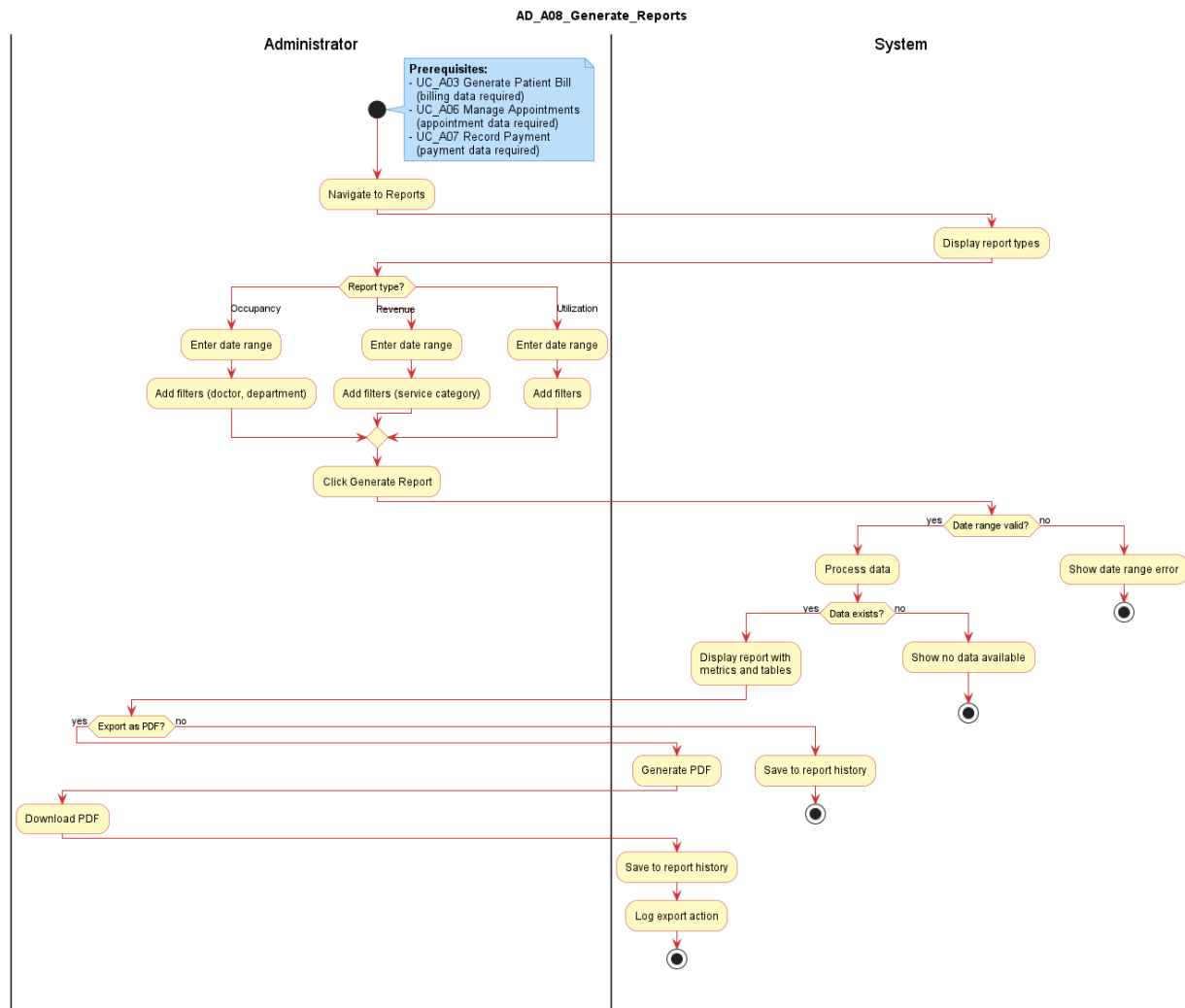
9.7 UC_A07: Record Payment

Activity diagram for UC_A07 – Record Payment



9.8 UC_A08: Generate Reports

Activity diagram for UC_A08 – Generate Reports

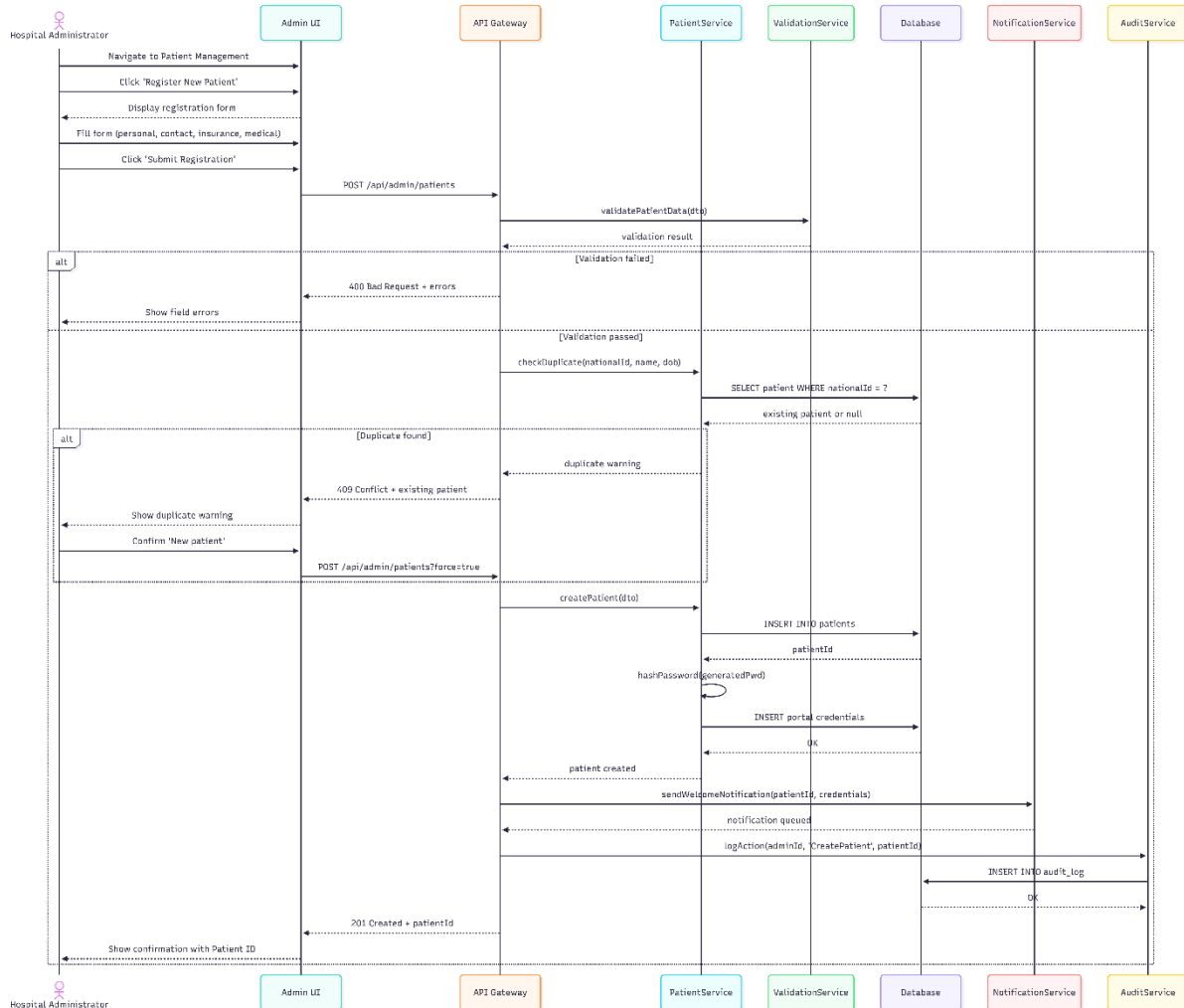


10. Sequence Diagrams

This section contains the sequence diagrams for each administrator use case. Each diagram illustrates the chronological interactions between the administrator, the UI layer, backend services, the database, and external components such as notification and audit services. Insert the corresponding PNG image in the placeholder below each title.

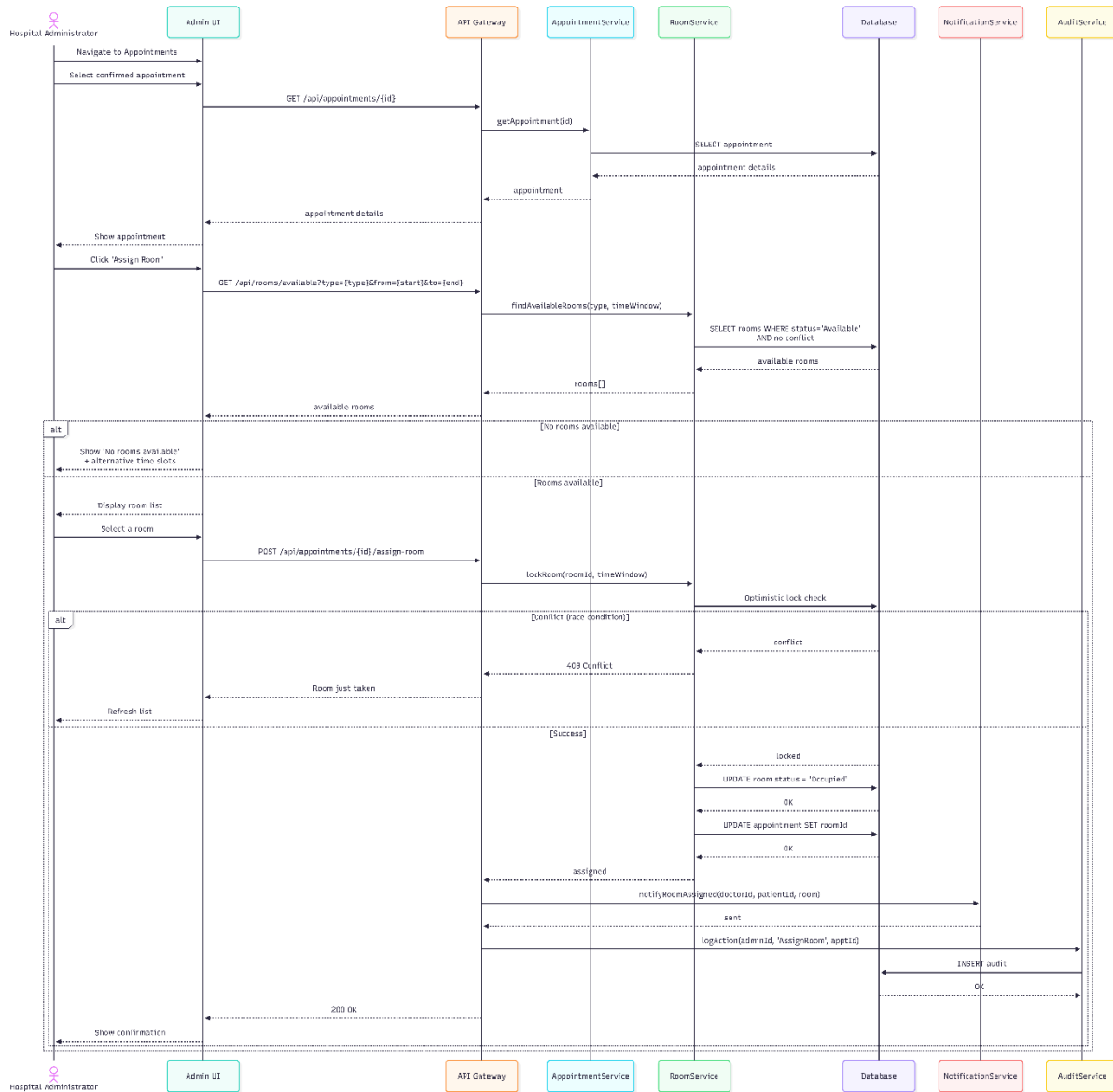
10.1 UC_A01: Register New Patient

Sequence diagram for UC_A01 – Register New Patient



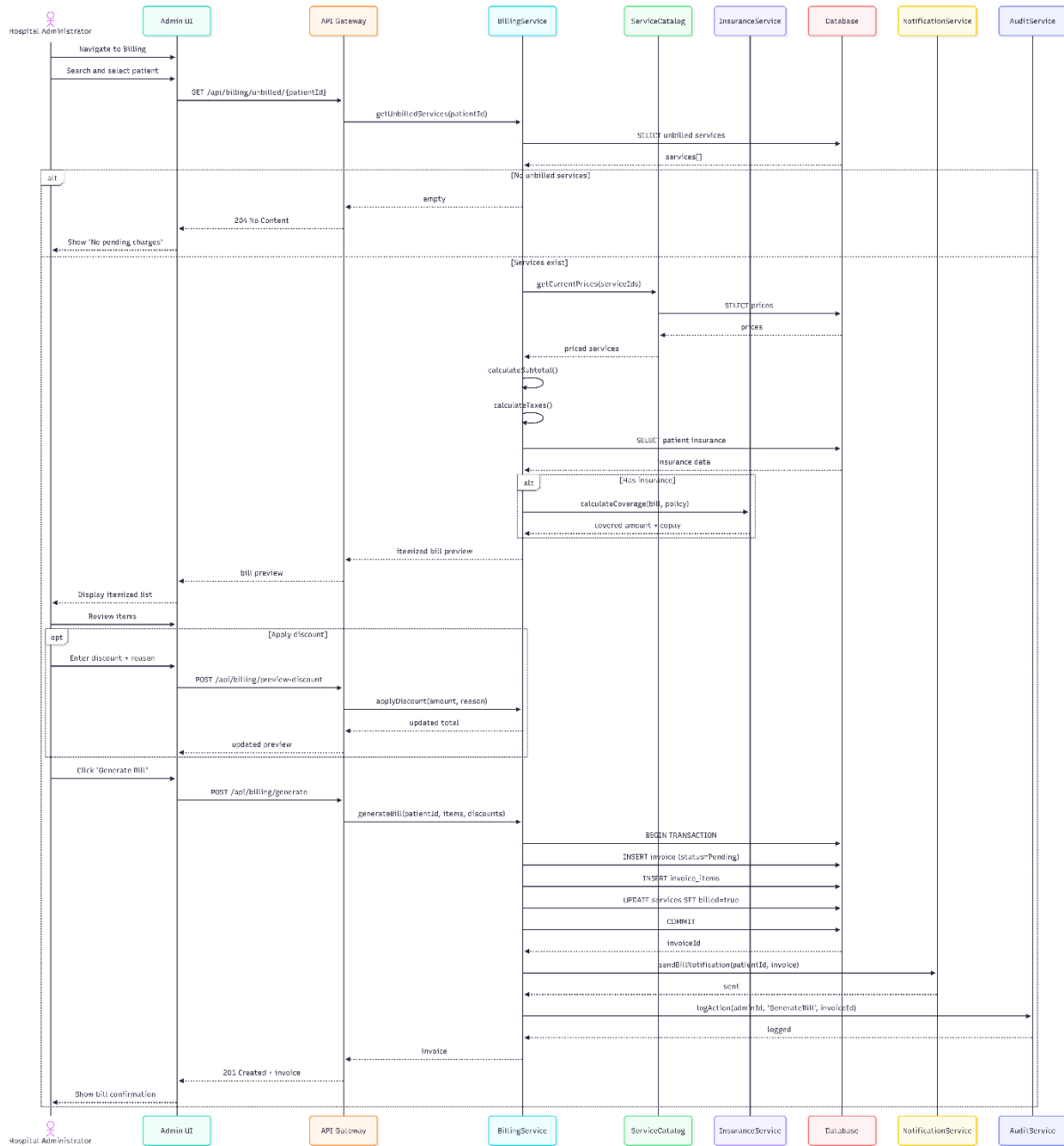
10.2 UC_A02: Assign Room to Appointment

Sequence diagram for UC_A02 – Assign Room to Appointment



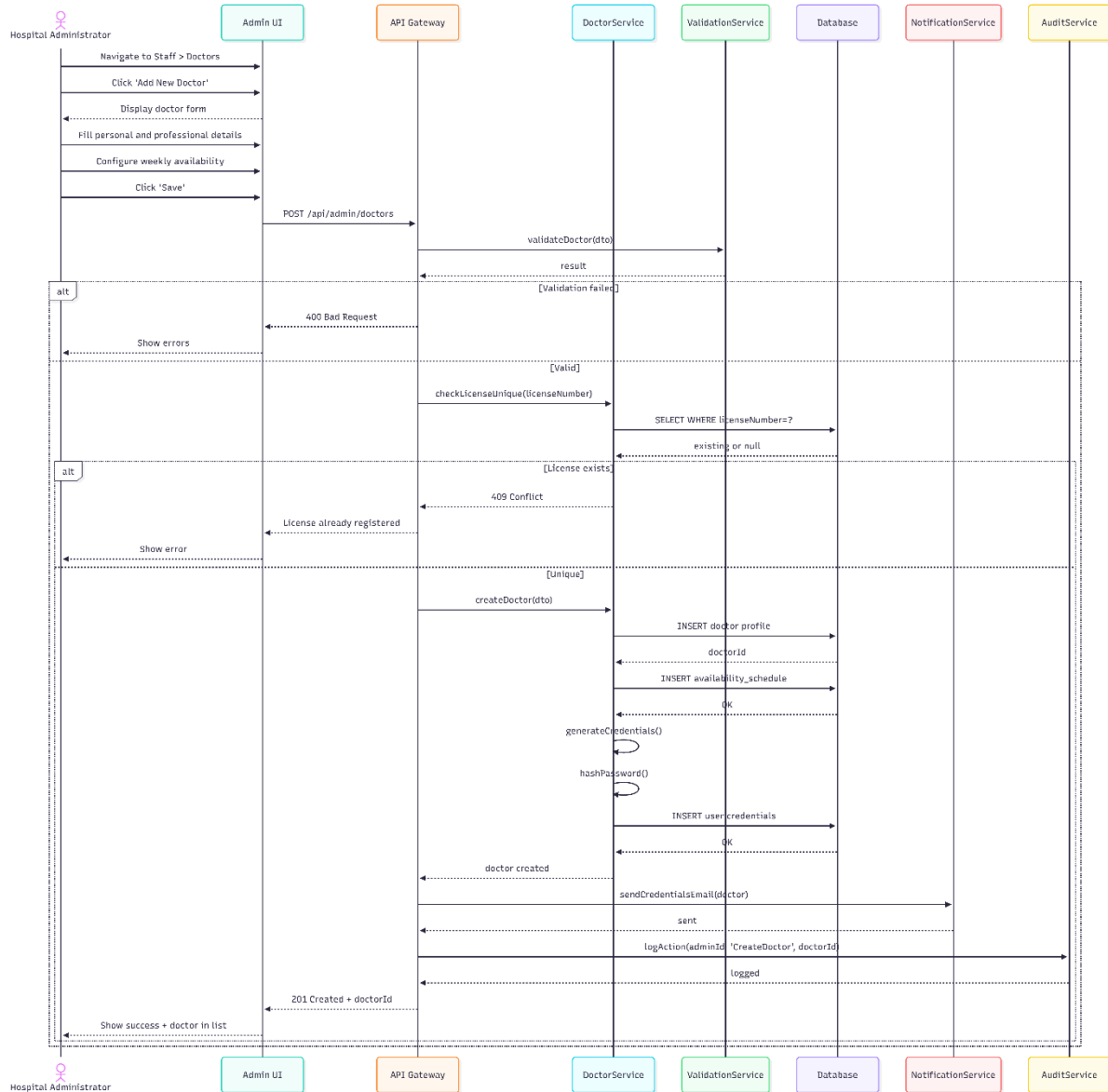
10.3 UC_A03: Generate Patient Bill

Sequence diagram for UC_A03 – Generate Patient Bill



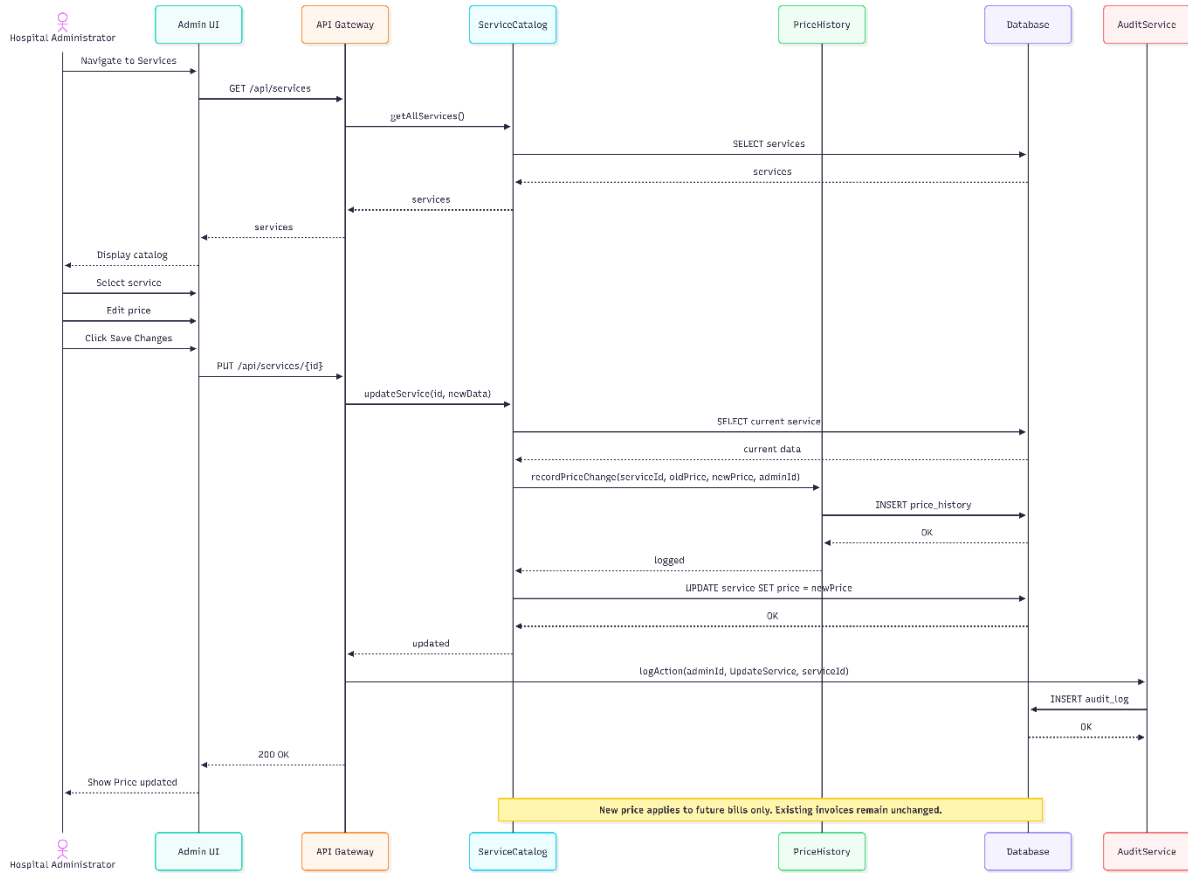
10.4 UC_A04: Manage Doctors (Add / Update / Deactivate)

Sequence diagram for UC_A04 – Manage Doctors (Add / Update / Deactivate)



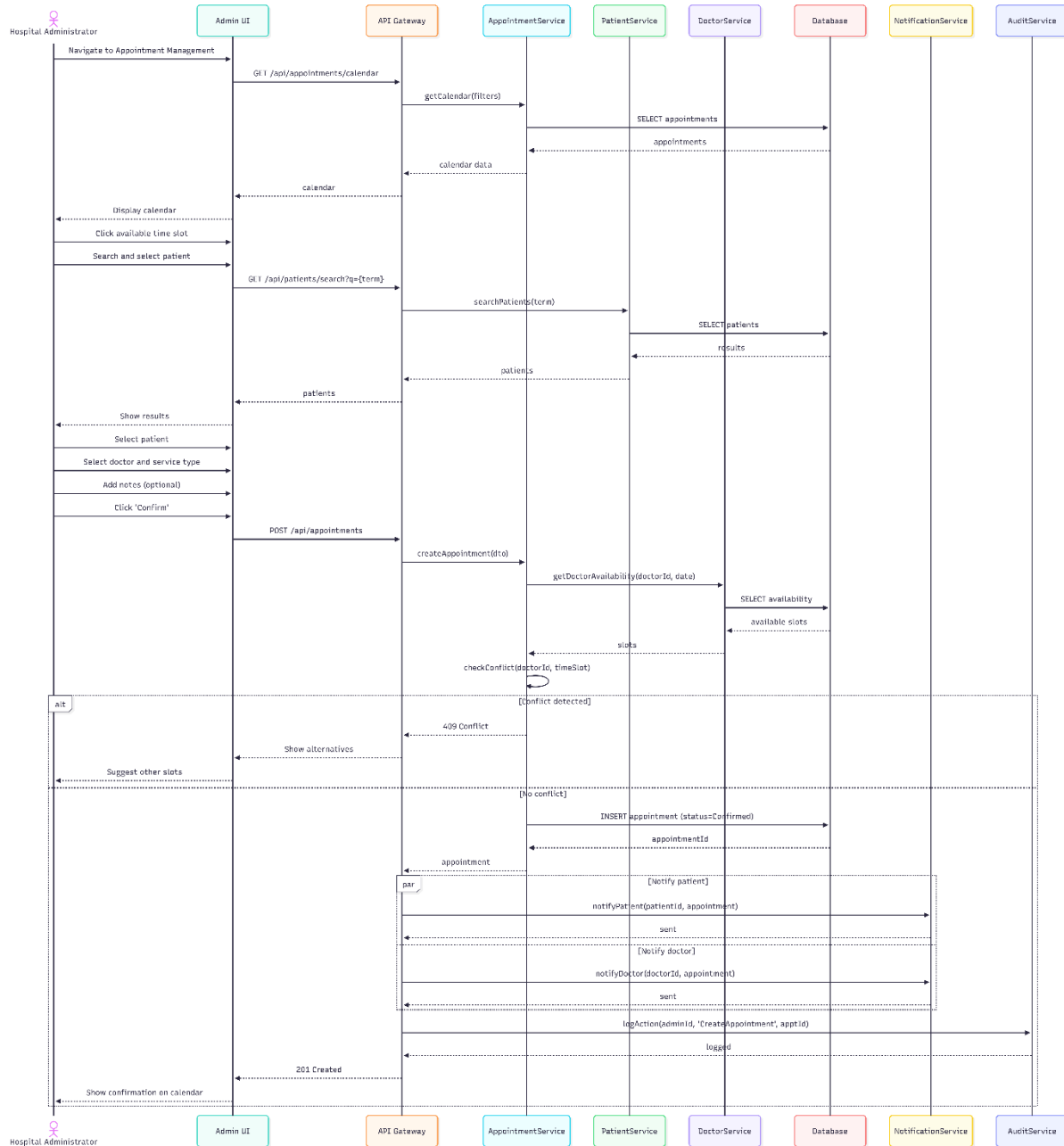
10.5 UC_A05: Manage Service Catalog and Prices

Sequence diagram for UC_A05 – Manage Service Catalog and Prices



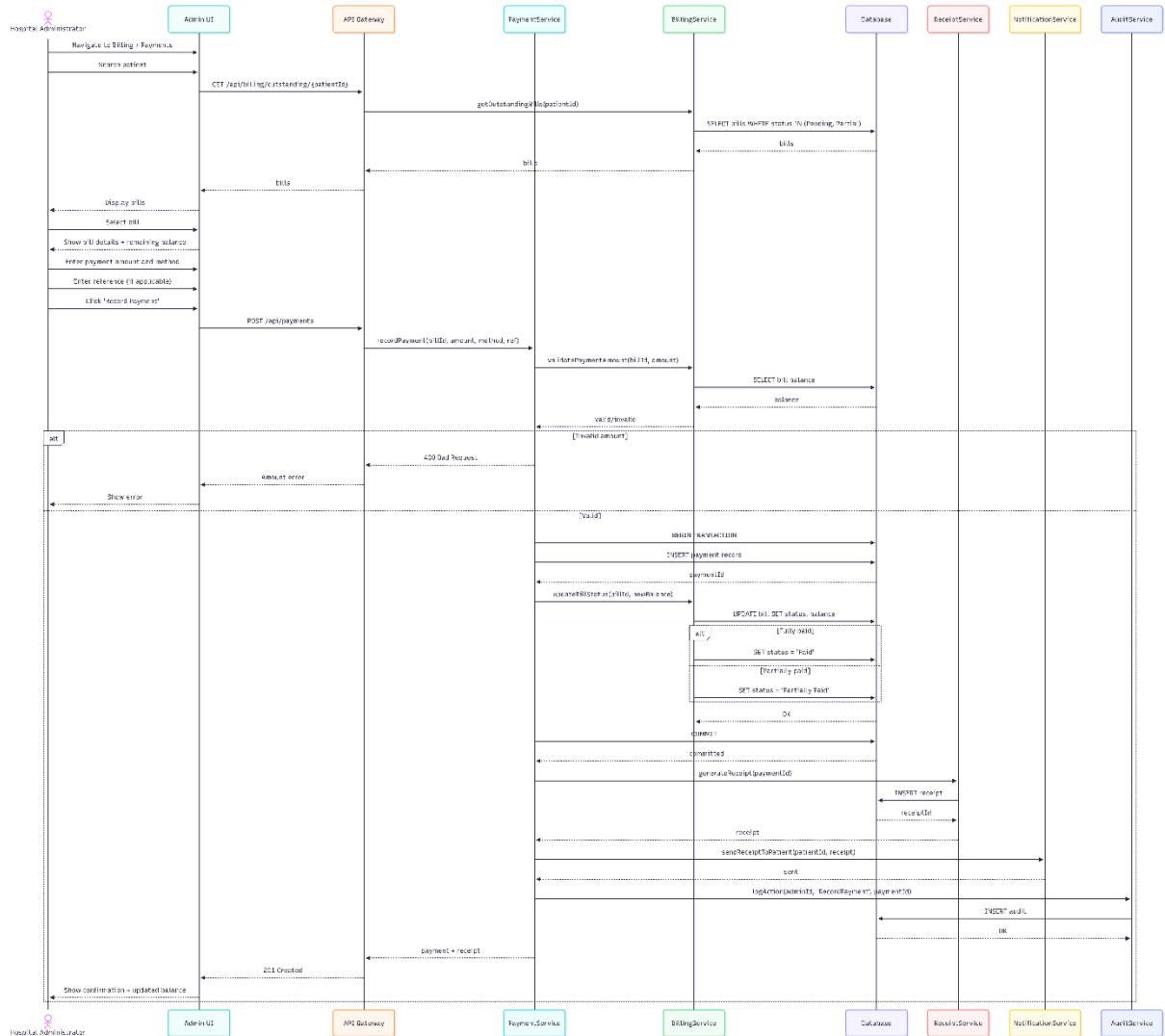
10.6 UC_A06: Manage Appointment Schedules

Sequence diagram for UC_A06 – Manage Appointment Schedules



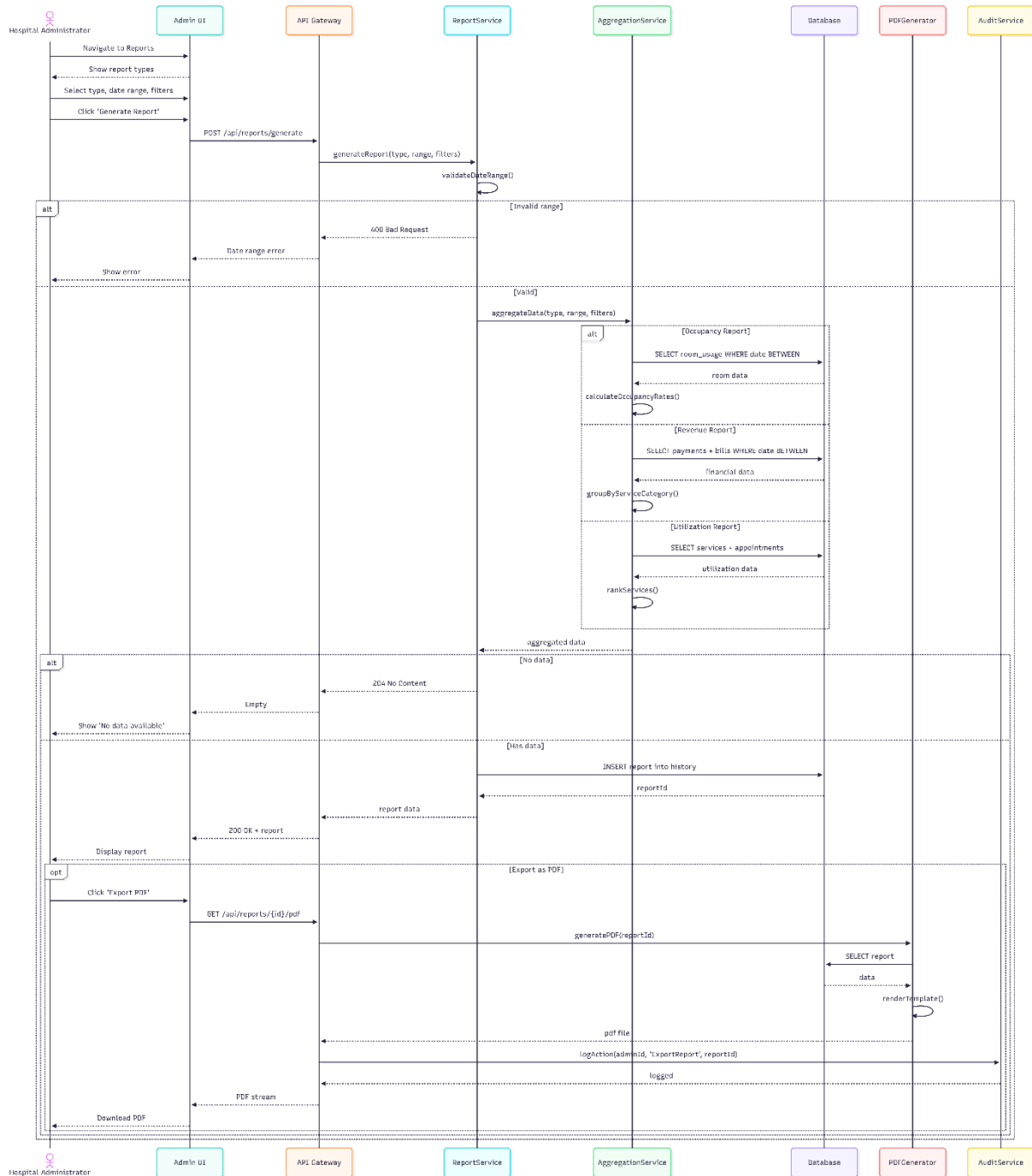
10.7 UC_A07: Record Payment

Sequence diagram for UC_A07 – Record Payment



10.8 UC_A08: Generate Reports

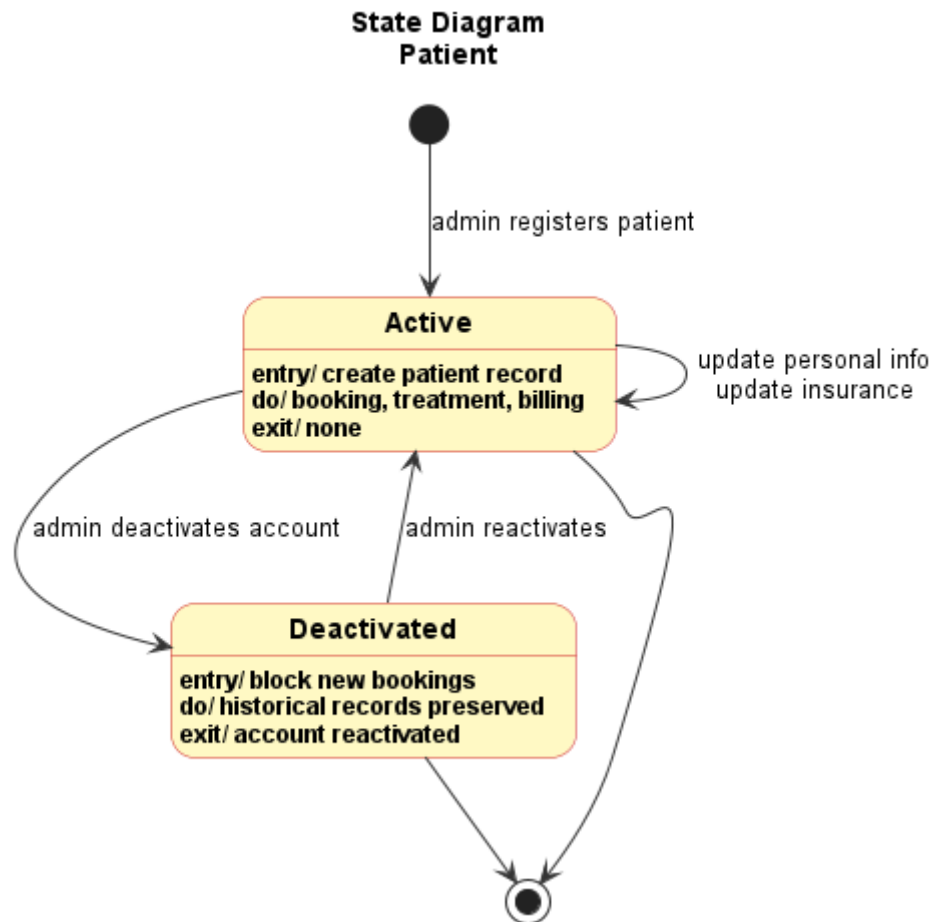
Sequence diagram for UC_A08 – Generate Reports



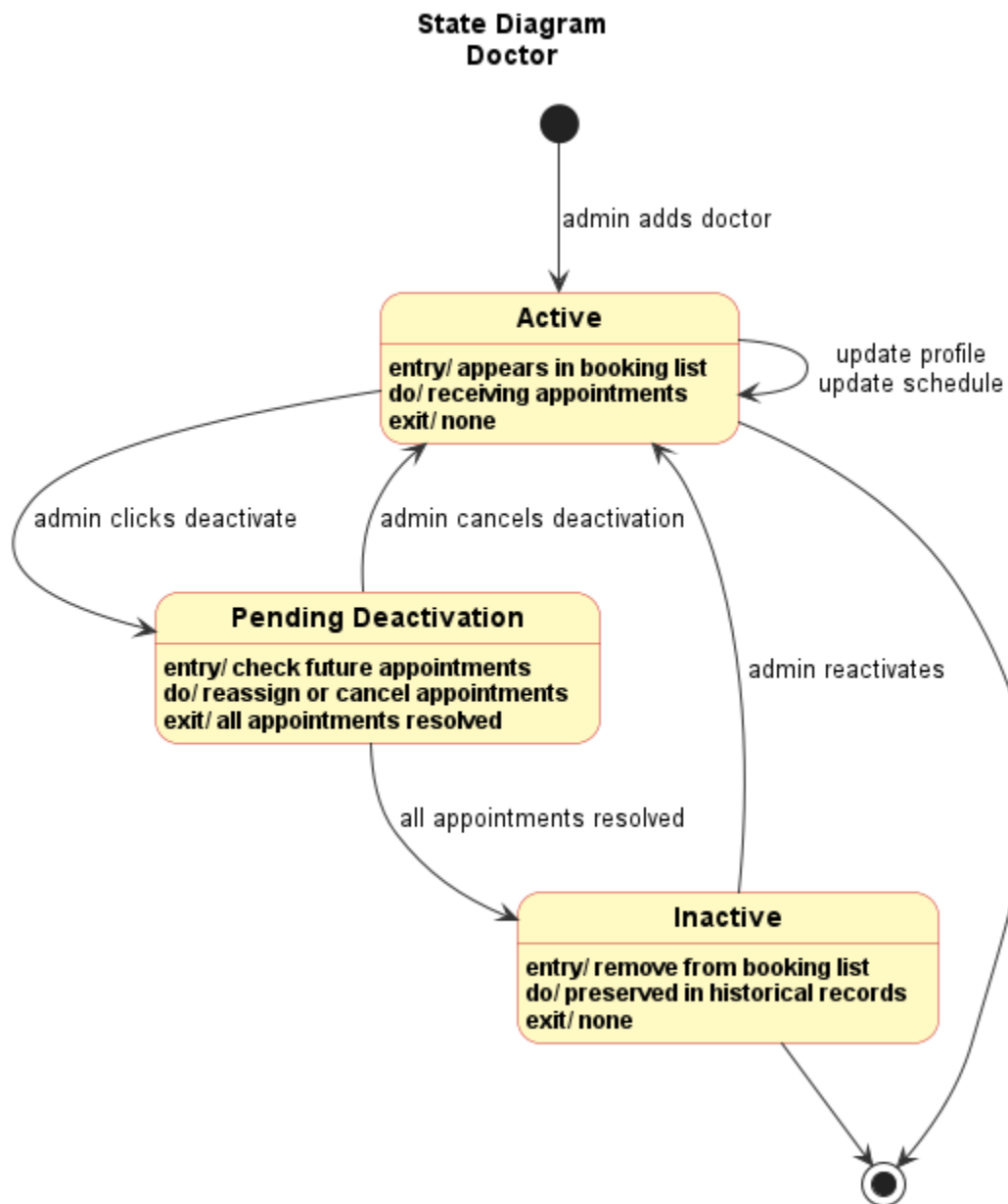
11. State Diagrams

This section contains the state diagrams for each entity managed by the administrator. Each diagram shows the lifecycle states, transitions, and entry/do/exit actions following UML state diagram notation.

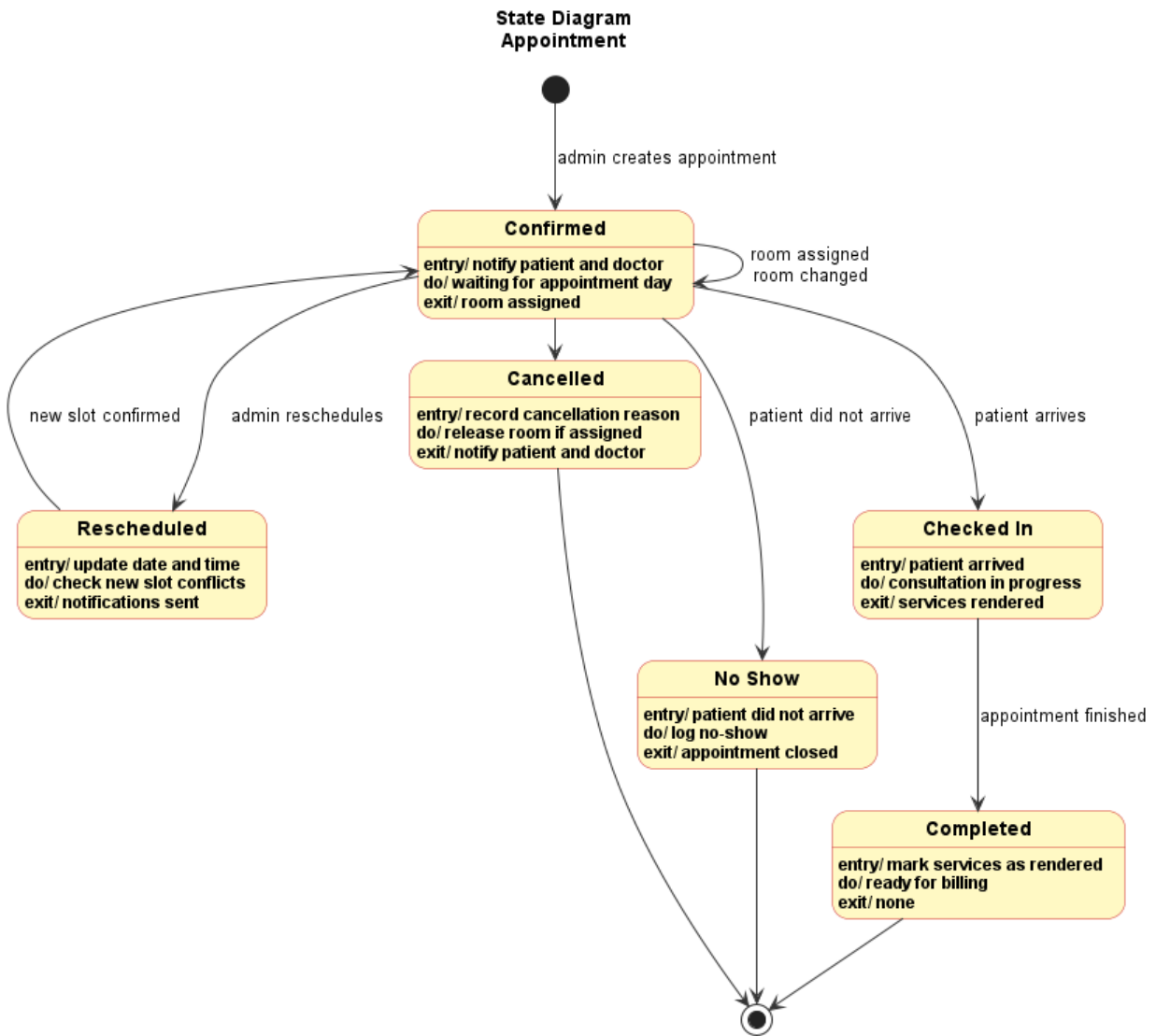
11.1 SD_Patient: Patient



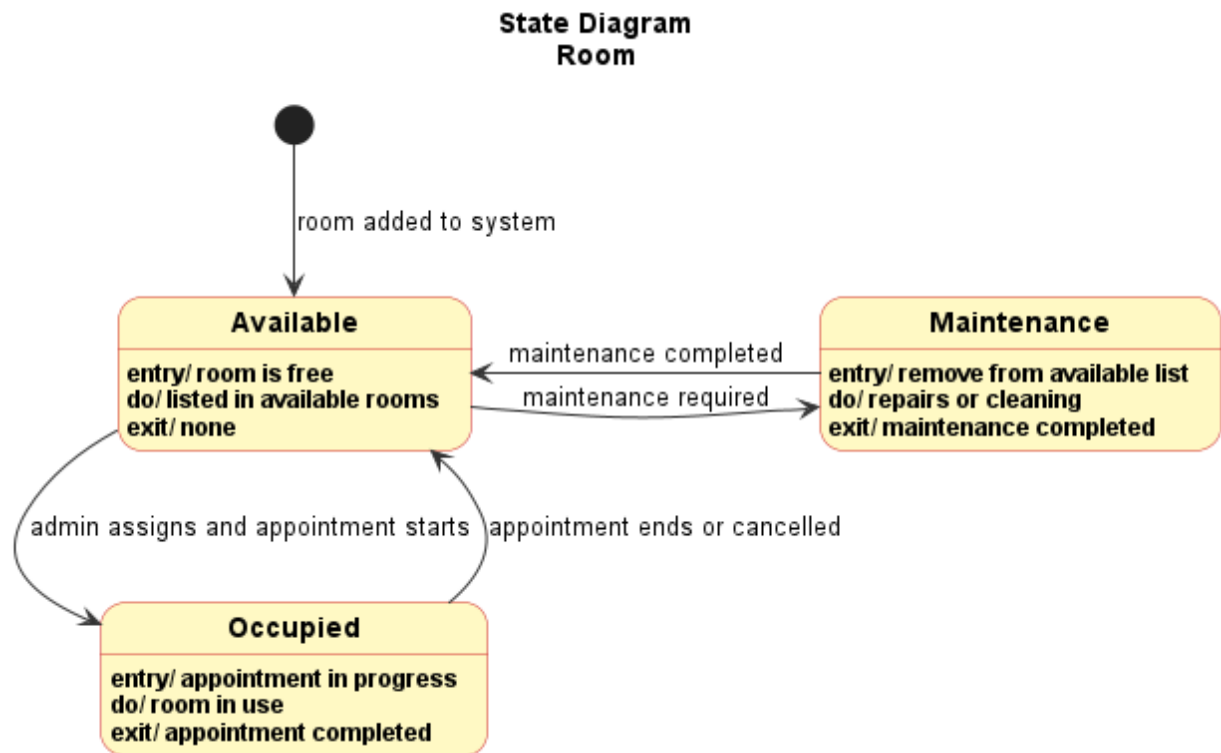
11.2 SD_Doctor: Doctor



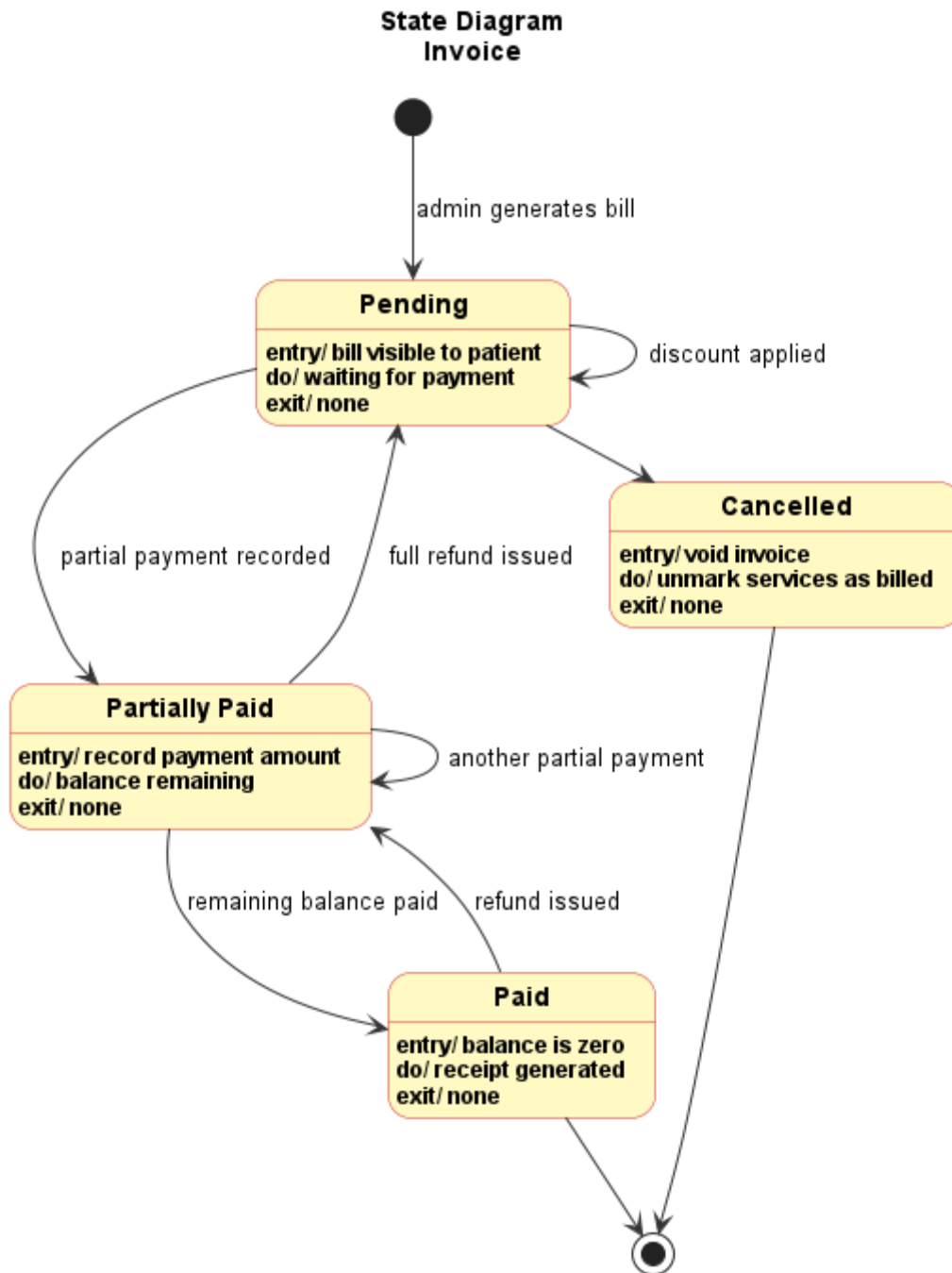
11.3 SD_Appointment: Appointment



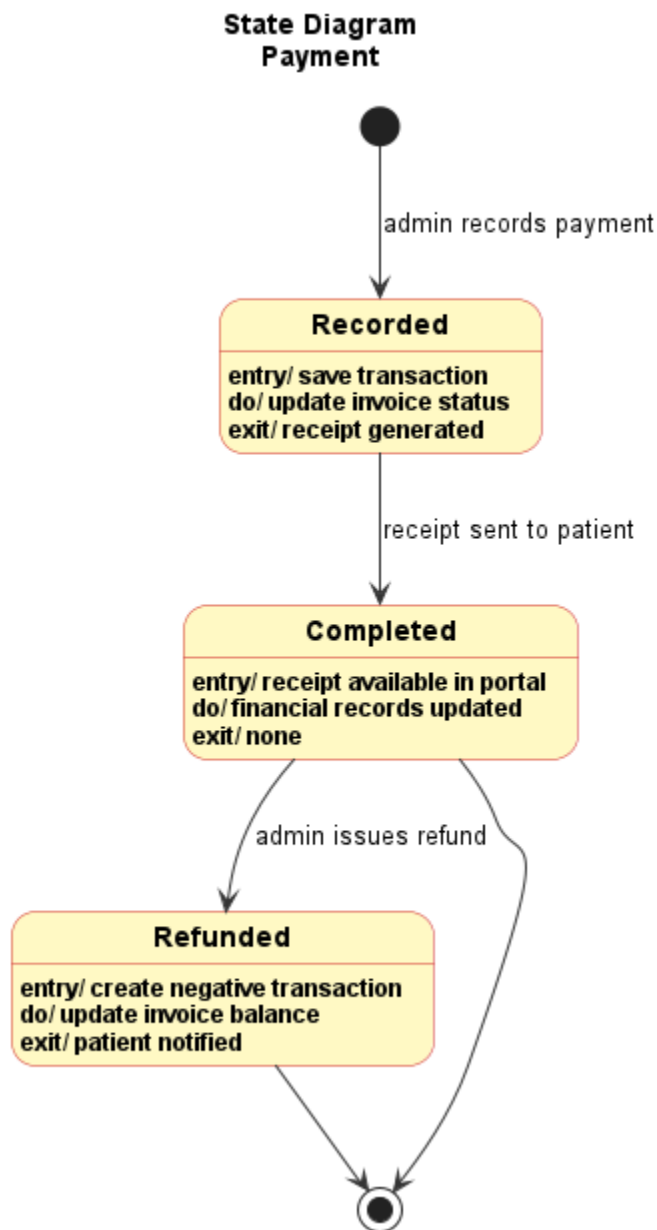
11.4 SD_Room: Room



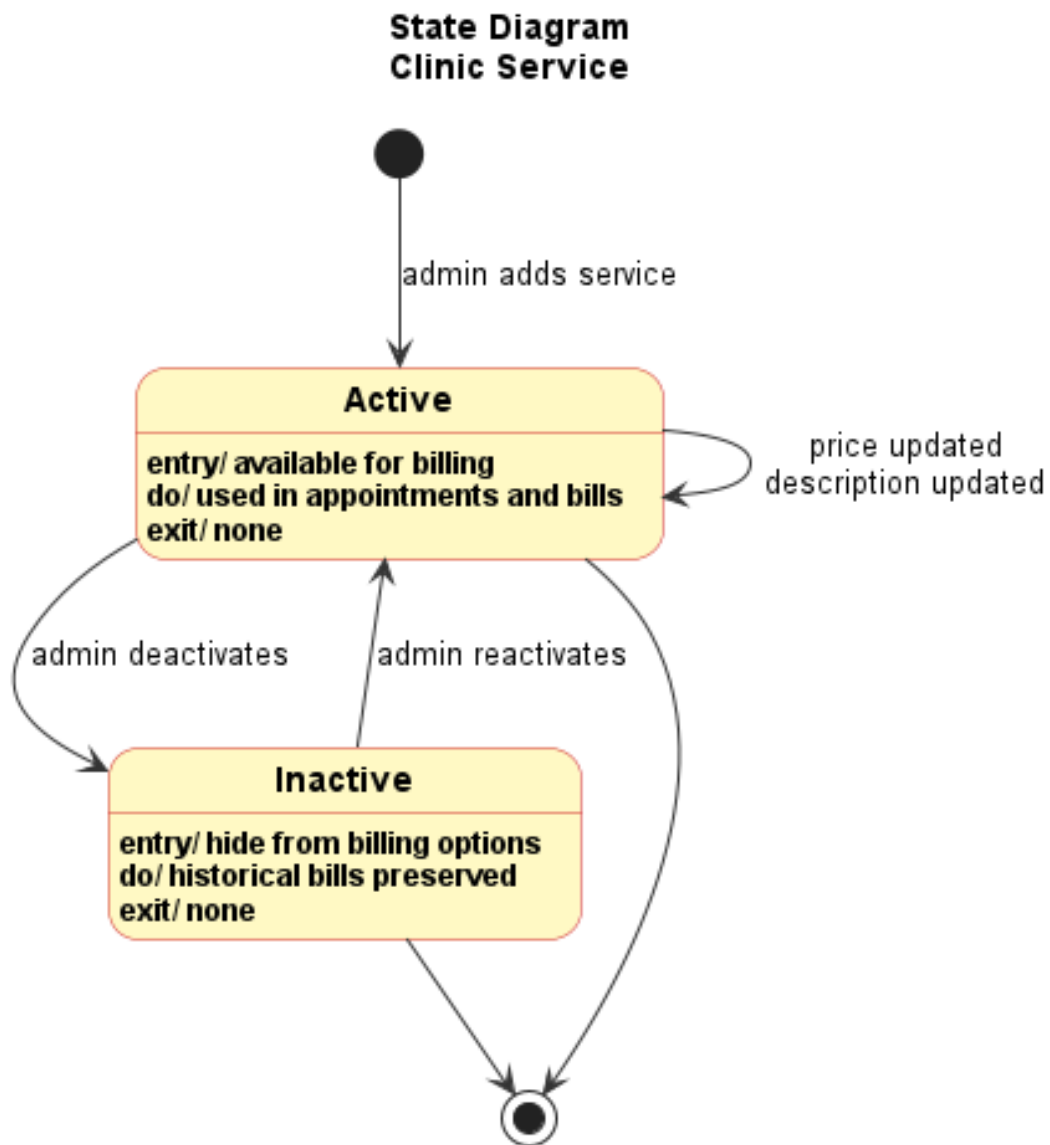
11.5 SD_Invoice: Invoice



11.6 SD_Payment: Payment



11.7 SD_ClinicService: Clinic Service



11.8 SD_AdminAccount: Admin Account

**State Diagram
Admin Account**

